



Multipole Expansion: Unifying Formalism for Earth and Planetary Gravitational Dynamics

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Abstract

The powerful method of multipole expansion has found wide utility in classical electromagnetics and quantum-mechanics. In contrast, the gravitational mechanics traditionally have only seen peripheral mentions of the corresponding (mass-density) multipoles in specific applications. In this paper, we develop the general theory of the multipole formalism for the classical two-body gravitational dynamics, a most prevalent subject in planetary (and global Earth) dynamics. We treat two general, well-encountered configurations: (i) the *interior* “mantle-inner core gravitational (MICG)” type of configuration consisting of concentric bodies; and (ii) the *exterior* type of configuration of “planet+satellite” separate bodies in the general form of tidal interactions. We derive concise and exact formulas in terms of multipole expansions; by retaining the leading term(s) of relevance, typically up to the quadrupolar terms that include the planetary triaxiality, one can evaluate to the precision desired while bearing in mind that higher subtleties are readily available in the higher multipole terms. The two-body problems that are so formulated range from potential energy, satellite orbit, to tides, librations of sorts and precession/nutation, to planetary rotational normal modes and wobbles. We demonstrate that, via its various symmetry properties, the multipole formalism provides a theoretical unification framework for these gravitational phenomena that are conventionally treated topic-by-topic in textbook literature.

Keywords Multipoles · Gravitation · Planet dynamics · Two-body problems · Tides · Mantle-core interactions

Article Highlights

- Gravitational multipole formalism for planetary two-body dynamics for energy, tides, precession/nutation, librations, rotational modes
- Two configurations are treated: *interior* “mantle-inner core gravitational” type, and the *exterior* “planet+satellite” type
- Multipole formalism provides a unification framework for gravitational phenomena conventionally treated topic-by-topic in literature

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1 Introduction

The powerful and elegant method of multipole expansion has found wide utility in theoretical physics, particularly in classical electromagnetics (EM) and quantum-mechanics (QM). The problems that are well treated by multipoles range from EM fields and radiations to atomic and molecular properties and interactions (e.g., Feynman et al. 1964; Landau and Lifshitz 1971; Jackson 1999).

The classical gravitational mechanics shares parallel physical fundamentals as EM in the sense of the conservative $1/r$ potential field, or the inverse-squared law force. However, traditionally it has only seen peripheral mentions of multipoles (or equivalent forms) in general (e.g., Snieder 2001; Chao and Gross 1987; Chao 2017a,b; Stirling 2017; Dehant and Mathews 2015), saving certain studies on general relativity and gravitational waves (e.g., Thorne 1980) and, in recent years, precise laboratory experiments such as testing the equivalence principle or determination of big G (e.g., Newman et al. 2014; Stirling and Schlamminger 2019). That has evidently been the case for planetary dynamics, arguably the most often encountered subject in classical gravitation. Yet, in planetary gravitation the multipoles are not really obscure: They appear in disguise as the familiar Stokes coefficients of the spherical-harmonic expansions of gravity field—the Stokes coefficients are actually normalized exterior multipoles (Chao and Gross 1987; see later), or as “inertia integrals” for arbitrary bodies for fast computation of Cartesian force and torque components (Ashenberg 2007; Hou et al. 2017; Boué 2017).

The present paper is motivated by the proposition: We already possess at our disposal a wealth of knowledge about the utility of multipoles in EM and QM, and, equally significant, about the powerful mathematical tool of (complex) spherical harmonics as developed in QM (e.g., Edmonds 1957). What would one be able to achieve by exploiting the multipole formalism to treat gravitational dynamics? We have in mind the *two-body* systems in particular that are most prevalent in planetary gravitation problems. The multipole formalism works as long as the two bodies do not have overlapping distance from the chosen coordinate origin; that is, no part of the “inner” body (closer to the origin) is more distant from the origin than any part of the “outer” body (farther from the origin). That is indeed satisfied in typical two-body planetary configurations, whereof we shall explore two types: (i) The *interior* “mantle-inner core gravitational (MICG)” type of configuration consisting of concentric bodies, analogous to the nucleus-electron Coulomb interactions within an atom; and (ii) the *exterior* type of configuration of “planet + satellite” separate bodies in the form of tidal interactions, analogous to the inter-atomic or inter-molecular EM interactions. The major difference between gravitation and EM in this regard is that in gravitation the monopole is typically the most prominent term, whereas the magnetic field has no monopole and the electricity is typically near-neutral with near-zero monopole. That potentially leads to unique, “surprise” consequences that are absent in EM and QM situations. In addition, we raise two *single-body* problems that can be treated conceptually via multipoles—gravitational inversion and gravitational self-energy.

The paradigm of the development of the multipole formalism for two-body gravitational dynamics is simple: (i) Deriving (“once and for all”) from first principles a general set of exact (and concise) equations in terms of multipole-multipole interactions. (ii) Performing onto the equation, if necessary, the appropriate coordinate transformations (translation and/or rotation) that suit the specific configuration of the problem at hand. (iii) Retaining the leading multipolar term(s) of relevance, to the desired precision. For demonstrations, in this paper as is often in conventional literature, we typically retain up to the quadrupolar terms that include

the planetary triaxiality, while bearing in mind that higher precisions with higher subtleties, as sometimes sought, are readily available in “the next terms”.

Besides the practical applications, we demonstrate, perhaps more significantly, that via its various symmetry properties the multipole formalism provides a theoretical conceptual unification framework for the variety of gravitational phenomena that are conventionally treated topic-by-topic in textbooks and in the literature. They range from potential energy, satellite orbit, to tides, librations of sorts and precession/nutation, to planetary rotational normal modes and wobbles. We state at the outset that our primary objective is to formulate up to the *forcings* that are responsible for the origin of the dynamic phenomenon. Thus we tend less to particularities of the *response* of the physical system to the forcings as time functions in most of the treated cases because (i) once formulated, they no longer pertain to the multipoles per se; (ii) they appear widely in the literature by way of solutions to the equations of motion.

2 Multipole Expansion of Gravitational Field

A planetary body can typically be well approximated by a near-spherical triaxial *quadrupole* body with the three principal moments of inertia $C > B > A$. The special case of a *monopole* spherical-symmetric body is characterized by $C = B = A$, whereas an axi-symmetric spheroidal body is characterized by $C > B = A$. A planet-centric (or for that matter the satellite-centric or geo-centric) coordinate system refers to the origin at the center of mass. The geographical coordinates are defined such that the z -axis is along the principal axis of the greatest moment of inertia (or the “ C axis”) which usually coincides with the spin axis. The other two principal axes, the “ B axis” and “ A axis” for the intermediate and least moments of inertia, respectively, then lie on the equatorial plane. In particular, the Earth’s geographical coordinates have the x -axis pointing to the Greenwich Meridian and the y -axis to the 90°E Meridian. For simplicity and without loss of generality, we can otherwise let them coincide with the B - and A -axes.

The Newtonian gravitational potential field $U(\mathbf{r})$ generated by a body of mass density $\rho(\mathbf{r}_0)$ and volume V_0 at field point $\mathbf{r}$ exterior to the body is given by the spatial convolution:

$$U(\mathbf{r}) = G \iiint_{V_0} \frac{\rho(\mathbf{r}_0)}{|\mathbf{r} - \mathbf{r}_0|} dV_0, \quad (1)$$

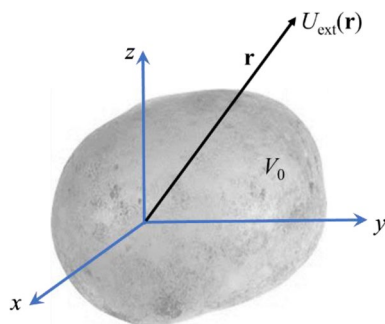
where $G = 6.6743 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ is the gravitational constant. The (positive) sign of U as defined is such that the gravitational acceleration is given by $\mathbf{g} = \nabla U$. Invoking the *addition theorem* leads to the following spherical-harmonic expansion for the Green’s function:

$$\frac{1}{|\mathbf{r}_1 - \mathbf{r}_2|} = \sum_{n=0}^{\infty} \sum_{m=-n}^n \frac{4\pi}{2n+1} \frac{r_1^n}{r_2^{n+1}} Y_{nm}(\Omega_1) Y_{nm}^*(\Omega_2) \quad (2)$$

(e.g., Morse and Feshbach 1953; Landau and Lifshitz 1971; Jackson 1999), where $\mathbf{r} = (\text{radius } r, \text{ co-latitude } \theta, \text{ East longitude } \lambda)$, and Ω is the solid angle representing θ and λ so that $d\Omega = \sin\theta \, d\theta \, d\lambda$ and $dV = r^2 \, dr \, d\Omega$, and $*$ denotes complex conjugation. The complex surface spherical harmonic function $Y_{nm}(\Omega) = (-1)^m \left(\frac{(2n+1)(n-m)!}{4\pi(n+m)!} \right)^{\frac{1}{2}} P_{nm}(\cos\theta) \exp(im\lambda)$ of degree n ($= 0, 1, \dots, \infty$) order m ($= -n, \dots, 0, \dots, n$), henceforth abbreviated as $[n, m]$, is fully normalized so that $\int Y_{nm'}(\Omega) Y_{nm}^*(\Omega) d\Omega = \delta_{nm'} \delta_{nm}$, where δ denotes the Kronecker delta function. Note that, while often omitted in geodesy literatures (e.g., Hofmann-Wellenhof and Moritz 2006; Stacey and Davis 2008), the Condon–Shortley phase factor

Table 1 Associated Legendre function $P_{nm}(\cos\theta)$ in spherical coordinates (co-latitude θ , east-longitude λ) up to degree $n=4$

	$m=0$	$m=1$	$m=2$	$m=3$	$m=4$
$n=0$	1				
$n=1$	$\cos\theta$	$\sin\theta$			
$n=2$	$(3\cos^2\theta - 1)/2$	$3\sin\theta\cos\theta$	$3\sin^2\theta$		
$n=3$	$(5\cos^3\theta - 3\cos\theta)/2$	$3/2\sin\theta(5\cos^2\theta - 1)$	$15\sin^2\theta\cos\theta$	$15\sin^3\theta$	
$n=4$	$(35\cos^4\theta - 30\cos^2\theta + 3)/8$	$5/2\sin\theta(7\cos^3\theta - 3\cos\theta)$	$15/2\sin^2\theta(7\cos^2\theta - 1)$	$105\sin^3\theta\cos\theta$	$105\sin^4\theta$

Fig. 1 The gravitational potential U_{ext} to the exterior of a gravitating body in the body-centric coordinates, where the exterior Q -multipole is appropriate

$(-1)^m$ is adopted here as commonly done in QM (as in Chao and Gross 1987) to facilitate the use of the angular momentum operator $\mathbf{L}$ (see later; also Arfken et al. 2012). The phase convention should be taken with care when referring to literature. The $P_{nm}(\mu) = \frac{(1-\mu^2)^{m/2}}{2^n n!} \frac{d^{n+m}}{d\mu^{n+m}} (\mu^2 - 1)^n$ is the $[n, m]$ th associated Legendre function; the low-degree ones are listed in Table 1 for reference. The symmetry property of P_{nm} is passed on to Y_{nm} , i.e., $Y_{n,-m} = (-1)^m Y_{nm}^*$; the factor $(-1)^m$ here is not the Condon–Shortley phase but originates from the definition of P_{nm} when applying Leibniz’s formula for the m th derivative of a product (e.g., Arfken et al. 2012).

2.1 Multipoles

Since the addition theorem Eq. (2) is symmetric with respect to $\mathbf{r}_1$ and $\mathbf{r}_2$, there are two ways Eq. (2) can be substituted into (1) depending on the assignment of suffixes 1 and 2. One way gives rise to the familiar form (subscript ext for exterior):

$$U_{\text{ext}}(\mathbf{r}) = 4\pi G \sum_{n=0}^{\infty} \sum_{m=-n}^n \frac{1}{(2n+1)r^{n+1}} \left[\iiint_{V_0} \rho(\mathbf{r}_0) r_0^n Y_{nm}(\Omega_0) dV_0 \right] Y_{nm}^*(\Omega), \quad (3)$$

that is valid for field points $r > r_0$ (see Fig. 1) and converges for $r \rightarrow \infty$. The other way gives rise to (subscript int for interior):

$$U_{\text{int}}(\mathbf{r}) = 4\pi G \sum_{n=0}^{\infty} \sum_{m=-n}^n \frac{r^n}{2n+1} \left[\iiint_{V_0} \rho(\mathbf{r}_0) r_0^{-(n+1)} Y_{nm}^*(\Omega_0) dV_0 \right] Y_{nm}(\Omega), \quad (4)$$

that is valid for field points $r < r_0$ (see Fig. 2) and converges for $r \rightarrow 0$.

The two respective complex-valued volume integrals in bracket in Eqs. (3) and (4) are:

$$Q_{nm} \equiv \iiint_{V_0} \rho(\mathbf{r}) r^n Y_{nm}(\Omega) dV = \iiint_{V_0} \rho(\mathbf{r}) R_{nm}(\mathbf{r}) dV, \quad (5a)$$

$$q_{nm} \equiv \iiint_{V_0} \rho(\mathbf{r}) r^{-(n+1)} Y_{nm}^*(\Omega) dV = \iiint_{V_0} \rho(\mathbf{r}) I_{nm}^*(\mathbf{r}) dV, \quad (5b)$$

where

$$R_{nm}(\mathbf{r}) \equiv r^n Y_{nm}(\Omega), \quad (6a)$$

$$I_{nm}(\mathbf{r}) \equiv r^{-(n+1)} Y_{nm}(\Omega), \quad (6b)$$

are, respectively, the *regular solid harmonic* and *irregular solid harmonic*, both being solution of the Laplace equation. Equations (3, 4) can then be written in shorthand as $U_{\text{ext}}(\mathbf{r}) = 4\pi G \sum_{n,m} Q_{nm} I_{nm}^*(\mathbf{r}) / (2n+1)$ and $U_{\text{int}}(\mathbf{r}) = 4\pi G \sum_{n,m} q_{nm} R_{nm}(\mathbf{r}) / (2n+1)$. Note that the choice of conjugation (*) placing (i.e., Y versus Y^*) in definition (5) is arbitrary owing to the symmetry of Eq. (2); our definition (5) follows that of Chao and Gross (1987), differing from Jackson (1999).

Equations (5a, 5b) are known as the $[n, m]$ th spherical-harmonic multipole moments of the mass density $\rho(\mathbf{r})$ (e.g., Chao and Gross 1987; Chao 2017a), henceforth the $[n, m]$ th *multipoles* for brevity, where n =degree and m =order. Specifically, Q_{nm} is the $[n, m]$ th *exterior* multipole, or Q -multipole, which is convergent as integration elements $r_0 \rightarrow 0$; q_{nm} is the $[n, m]$ th *interior* multipole, or q -multipoles, which is convergent as $r_0 \rightarrow \infty$.

The multipolar components of the two types have the radial dependence of $r^{-(n+1)}$ and r^n , respectively. These radial dependences are the transfer functions to be applied to the process of (global) downward or upward continuation of gravity field, during which the multipoles themselves remain (of course) as constant entities. The pair of multipoles of the same n differ in dimension by r^{2n+1} . There are $2n+1$ multipoles of either type of different m 's to a given n . Only $n+1$ multipoles of non-negative m 's are independent, given their symmetry property $Q_{-n-m} = (-1)^m Q_{nm}^*$, $q_{n-m} = (-1)^m q_{nm}^*$ by virtue of $Y_{-n-m} = (-1)^m Y_{nm}^*$.

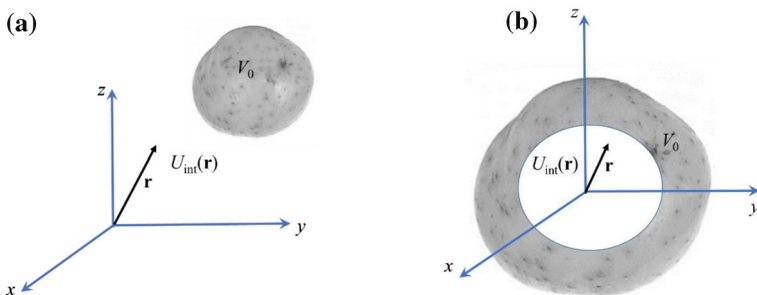


Fig. 2 The gravitational potential U_{int} of a gravitating body such that the coordinate origin is “interior” to the body, where the interior q -multipole is appropriate. **a** A “satellite” in planet-centric coordinates. **b** A “core” cavity inside a planet

Mathematically the expression (5) has the form of an inner product, i.e., the product of two functions integrated over the entire domain of the functions. Just like the Fourier transform is the inner product of a function with a sinusoidal function (as kernel), or Hankel transform that with a (weighted) Bessel kernel, here a multipole is an (unnamed) transform of the density function with respect to the solid-harmonic kernel. It transforms the density function from the 3-D spatial domain (specifically the volume V_0 where ρ is non-zero) to the $[n, m]$ “spectral” domain. In this sense, the multipoles represent the spectral breakdown of the density distribution with larger n representing higher spatial subtleties. The multipole expansions (3, 4) then represent the corresponding inverse transform analogous to the Fourier series.

Thus, Q_{nm} is the inner product, or “projection”, of $\rho(\mathbf{r})$ onto R_{nm} that is appropriate for integration over a solid body in a body-centric coordinates as depicted in Fig. 1. q_{nm} is the projection of $\rho(\mathbf{r})$ onto I_{nm}^* that is appropriate for integration over a solid body when the coordinate origin is set outside the body, as depicted in Fig. 2 for two relevant cases—a satellite in a planet-centric coordinates, and the interior space such as a core cavity chamber inside a planet. On account of their radial weightings, as n increases the Q -multipoles become increasingly more sensitive toward the “outer” parts of the body that are farther away from the origin, whereas the q -multipoles become increasingly more sensitive toward the “inner” parts of the body that are closer to the origin.

2.2 Exterior Multipoles Versus Stokes Coefficients of a Planet

In planetary geodesy, the spherical-harmonic expansion of the external gravitational field U_{ext} is customarily expressed in terms of the *Stokes coefficients* C_{nm} and S_{nm} :

$$U_{\text{ext}}(\mathbf{r}) = \frac{GM}{r} \sum_{n=0}^{\infty} \sum_{m=0}^n \left(\frac{R}{r}\right)^n \tilde{P}_{nm}(\cos\theta) (C_{nm} \cos m\lambda + S_{nm} \sin m\lambda) \quad (7)$$

in reference to the planet’s mean radius R and mass M (Kaula 1966; Torge 1989), where $\tilde{P}_{nm}(\cos\theta) = [(2 - \delta_{m0})(2n+1)(n-m)!/(n+m)!]^{1/2} P_{nm}(\cos\theta)$ is the 4π -normalized associated Legendre function. Equating Eqs. (3) and (7), one finds:

$$C_{nm} + iS_{nm} = \frac{(-1)^m 2\sqrt{(2 - \delta_{m0})\pi}}{(2n+1)MR^n} Q_{nm}, \quad m = 0, 1, 2, \dots, n \quad (8)$$

i.e., the (dimensionless) Stokes coefficients are just normalized Q -multipoles of the planet. They constitute the physical observables directly from external gravitational measurements. We stress that the formalism in term of the Stokes coefficients as two separate *real*-valued quantities fails to embrace the mathematical efficacy of the complex spherical harmonics (as does QM), especially when motions are involved as in this paper.

The naming practice for specific multipoles is according to 2^n (see Table 2), whereby the total general potential U can be symbolically partitioned into $U_{\text{monopole}} + U_{\text{dipoles}} + U_{\text{quadrupoles}} + U_{\text{octupoles}} + U_{\text{higher-multipoles}}$ as follows. The $n=0$ Q -monopole is a true scalar: $Q_{00} = \sqrt{1/4\pi} C_{00} M = \sqrt{1/4\pi} M$ signifies the total mass that is independent of the coordinate ($C_{00} = 1$ identically by the normalization). The monopole component of U_{ext} (Eq. 3) is thus GM/R , as expected. It is readily seen (from the orthogonality of spherical harmonics) that all multipoles but the monopoles vanish in the case of spherical symmetry.

The $n=1$ Q -dipoles are $Q_{10} = \sqrt{3/4\pi} M z_1$, $Q_{11} = -\sqrt{3/8\pi} M(x_1 + iy_1)$, whereof the vector $\langle x_1, y_1, z_1 \rangle = \sqrt{3} R \langle C_{11}, S_{11}, C_{10} \rangle$ in the Cartesian coordinates locates the

Table 2 The degree of deficiency as a function of spherical-harmonic degree n in the 3-D gravitational inversion (adopted from Chao 2005)

Degree n	# multipoles $(2n + 1)$	# moments $(n + 1)(n + 2)/2$	Degree of deficiency $n(n-1)/2$
0	1 (monopole)	1 (total mass)	0
1	3 (dipole)	3 (center of mass)	0
2	5 (quadrupole)	6 (inertia tensor)	1
3	7 (octupole)	10 (3rd moment)	3
4	9	15	6
5	11	21	10
6	13	28	15
100	201	5151	4950

body's center of mass. Conversely, the Q -dipoles vanish identically when the coordinate origin is set at the center of mass, as is typically the case.

Much about the planetary gravitation pertains to $n=2$ Q -quadrupoles (e.g., Bullen 1975; Stacey and Davis 2008) which are closely related to the inertia tensor elements (for detailed derivation see Chao and Gross 1987). The dominant term for the Earth next to its monopole is the oblateness due to the centrifugal force of Earth's spin, embodied in the zonal $Q_{20} = -\sqrt{5/4\pi} J_2 MR^2$ in terms of $J_2 = -\sqrt{5} C_{20} = [C - (A + B)/2]/MR^2$. The sectoral $Q_{22} = \sqrt{15/32\pi}(B - A)\exp(i2\Lambda_0)$ embodies the equatorial ellipticity $B - A$ (about $\sim 1/150$ of $C - A$ for the Earth), where the angle Λ_0 ($= -14.93^\circ$ for the Earth) is the East longitude of the A axis in the geographical coordinates (Liu and Chao 1991; Chao and Yu 2020). The tesseral $Q_{21} = \sqrt{15/8\pi}(K_{zx} + iK_{yz})$ (Chao and Gross 1987) relates to the zx and yz inertia product elements of the inertia tensor $\mathbf{K}$; its change excites the polar motion, the departure of the Poles from the C axis. It vanishes if the z -axis coincides with the C axis as is often so stipulated. Aside from $C_{00} = 1$ and $C_{20} \sim O(10^{-3})$, the degree-variance spectrum of higher-degree Stokes coefficients of the Earth follows the empirical Kaula rule of thumb (Kaula 1966): $\Sigma_m (C_{nm}^2 + S_{nm}^2)/(2n + 1) \approx 10^{-10}/n^4$; accordingly C_{nm}, S_{nm} are no more than $\sim O(10^{-6})$ and diminish in magnitude with n .

By contrast, the q -multipoles find no direct physical counterparts as do the low-degree Q -multipoles (cf. Stirling and Schlamminger 2019). In particular, the monopole q_{00} remains invariant with respect to coordinate rotation but is radially dependent. For example, a spherical shell (configuration of the type of Fig. 2b) of constant density ρ with the inner and outer radii R_1 and R_2 has $q_{00} = \sqrt{\pi} \rho (R_2^2 - R_1^2)$ and vanishing higher multipoles (due to spherical symmetry). The corresponding monopolar field $U_{\text{int}} = 2\pi G \rho (R_2^2 - R_1^2)$ (r^0 term in Eq. 4) inside the spherical shell is uniform and position-independent, a well-known fact found in textbooks, and reduces to the familiar expression of constant $U_{\text{int}} = GM/R$ if the shell thickness is infinitesimal, as expected.

For a body of uniform density ρ_0 , the multipoles are solely dependent on its geometrical shape. If we express the (singly-connected) topography in spherical harmonics as $r(\Omega) = R \left[1 + \sum_{n=1}^{\infty} \sum_{m=-n}^n H_{nm} \tilde{Y}_{nm}(\Omega) \right]$ referenced to the average radius $R = \int R(\Omega) d\Omega / 4\pi$ (i.e., the “monopole” for r) similarly as Eq. (7) (with 4π -normalization), and substitute it into Eq. (5a), then for such a body one gets (for detail see Chao and Rubincam 1989): $C_{nm} + iS_{nm} = \frac{R^3}{(2n+1)V} [4\pi H_{nm} + l.c.(\text{double-products of } H_{st}) + l.c.(\text{triple-products of } H_{st}) + \dots]$ where $l.c.(\dots)$ means linear combination. This relation has found relevance in evaluating

the low-degree gravitational field of the odd-shaped Phobos and Deimos satellites of Mars (Chao and Rubincam 1989; Rubincam et al. 1995). For a near-spherical body with $H_{n \neq 0, m} \ll 1$, neglecting all but the first term yields $C_{nm} + iS_{nm} \approx [3/(2n+1)] H_{nm}$. In particular, $n=1$ indicates that the center of mass coincides with the center of figure, rightfully for a uniform-density body. For $[2, 0]$, the gravitational triaxiality equals to $3/5$ of the figure triaxiality. Higher-degree multipoles diminish following $(2n+1)^{-1}$ relative to the corresponding subtleties in shape.

The multipole formalism applies regardless of the state of the body, e.g., whether the body is rigid or non-rigid, and equally well to *time-variable* gravity when the mass distribution of the body changes with time, only that the multipoles (hence Stokes coefficients) now become functions of time. Thus, by Eq. (5a) the change (indicated by Δ) of Q -multipole under an infinitesimal perturbation can be written as

$$\Delta Q_{nm}(t) = \int \Delta \rho(\mathbf{r}; t) r^n Y_{nm}(\Omega) dV, \quad (9a)$$

in the Eulerian description (e.g., Chao et al. 1987), or:

$$\Delta Q_{nm}(t) = \int \rho(\mathbf{r}) r^{n-1} \mathbf{u}(\mathbf{r}, t) \cdot [n \hat{\mathbf{r}} Y_{nm}(\Omega) + \nabla_s Y_{nm}(\Omega)] dV, \quad (9b)$$

in the Lagrangian description (Chao and Gross 1987), where $\mathbf{u}$ is the infinitesimal displacement field, $\hat{\mathbf{r}}$ denotes unit vector, $\nabla_s = \hat{\boldsymbol{\theta}} \partial_\theta + \hat{\boldsymbol{\lambda}} \csc \theta \partial_\lambda$ is the surface gradient operator so that $\nabla = \hat{\mathbf{r}} \partial_r + (1/r) \nabla_s$, and the integrations are over the instantaneous entire volume. The two descriptions are equivalent to first order (e.g., Backus et al. 1996) under the conservation of mass; the choice depends on the specific application at hand. The Eulerian approach is fitting when the mass redistribution $\Delta \rho$ is specified, such as for the hydrological cycle on Earth surface, whereas the Lagrangian approach is useful when the displacement field $\mathbf{u}$ is provided (or modeled) such as for earthquake faulting.

The satellite missions of GRACE (Tapley et al. 2004) and follow-ons observe the variation in the geoid (the gravitational equipotential surface closest to the Earth spheroidal figure) which by Bruns' formula can be written as $\Delta N(\mathbf{r}) = R \sum_{n,m} \tilde{P}_{nm}(\cos \theta) (\Delta C_{nm} \cos m\lambda + \Delta S_{nm} \sin m\lambda)$ from Eq. (7), where the summation over n starts at $n=2$ in the geocentric coordinates. The corresponding gravity anomaly field, via pertinent spatial gradient (e.g., Torge 1989), becomes $\Delta g(\mathbf{r}) = (GM/R^2) \sum_{n,m} (n-1)(R/r)^{n+2} \tilde{P}_{nm}(\cos \theta) (\Delta C_{nm} \cos m\lambda + \Delta S_{nm} \sin m\lambda)$, whereof the weighting of each harmonic term is magnified by the factor $(n-1)$.

Furthermore, a convenient GRACE product known as the “equivalent water thickness” field (e.g., Wahr et al. 1998) or surface “mascon” solution (e.g., JPL GRACE Tellus website) has the expression

$$\Delta \sigma(\theta, \lambda) = (R \rho_{ave}/3) \sum_{n,m} [(2n+1)/(1+k'_n)] \tilde{P}_{nm}(\cos \theta) (\Delta C_{nm} \cos m\lambda + \Delta S_{nm} \sin m\lambda),$$

whereof each harmonic term is magnified by the factor $(2n+1)$ and further augmented by the factor $1/(1+k'_n)$ to “undo” the elastic loading effect (here ρ_{ave} is the Earth's mean density, k'_n its load Love number of degree n). This relation is valid and has physical meaning when and only when all mass variations that produce the time-variable gravity (in the form of $\Delta \rho$ or $\mathbf{u}$ in Eq. 9) occur on the (2-D) Earth surface, because only then will the gravitational inversion become unique (Chao 2005, 2016; see Sect. 3).

2.3 Analogy in Planetary Magnetic Field

By convention, the geomagnetic scalar potential field V (for which the magnetic induction field is given by $\mathbf{B} = -\nabla V$) in the exterior of Earth in the absence of electric current (hence satisfying Laplace equation) is expressed as:

$$V_{\text{ext}}(\mathbf{r}) = R \sum_{n=1}^{\infty} \sum_{m=0}^n \left(\frac{R}{r}\right)^{n+1} \bar{P}_{nm}(\cos \theta) (g_{nm} \cos m\lambda + h_{nm} \sin m\lambda) \quad (10)$$

for the internal-generated field, or

$$V_{\text{int}}(\mathbf{r}) = R \sum_{n=0}^{\infty} \sum_{m=0}^n \left(\frac{r}{R}\right)^n \bar{P}_{nm}(\cos \theta) (g'_{nm} \cos m\lambda + h'_{nm} \sin m\lambda) \quad (11)$$

for the external-generated field. For the record, the normalization adopted here is that of the Schmidt quasi-normalized associated Legendre function $\bar{P}_{nm} = [(2 - \delta_{m0})(n - m)!/(n + m)!]^{1/2} P_{nm}$ (and correspondingly for $\bar{Y}_{nm}$), different than Eq. (7). Known as the Gauss coefficients of the geomagnetic field, g_{nm} and h_{nm} have the unit of tesla (in contrast to the dimensionless Stokes coefficients).

Just as mass density ρ is the source of gravitation, the electric current density vector $\mathbf{J}$ in the form of the scalar $(\mathbf{r} \cdot \nabla \times \mathbf{J})$ in volume V_0 where $\mathbf{J}$ is non-zero (e.g., the geodynamo for the Earth) constitutes the geomagnetic source (Gray 1979; Backus et al. 1996; Lowes and Duka 2011). Thus, corresponding to Eq. (5a) and (5b), the following integrals are identified to be the *magnetic* exterior Q -multipoles and interior q -multipoles, respectively:

$$Q_{nm} = \iiint_{V_0} (\mathbf{r} \cdot \nabla \times \mathbf{J}) r^n \bar{Y}_{nm}(\Omega) dV, \quad q_{nm} = \iiint_{V_0} (\mathbf{r} \cdot \nabla \times \mathbf{J}) r^{-(n+1)} \bar{Y}_{nm}^*(\Omega) dV. \quad (12)$$

Accordingly and corresponding to Eq. (8), the Gauss coefficients are related by

$$g_{nm} + ih_{nm} = \frac{\mu_0}{4\pi(n+1)(2n+1)} \frac{1}{R^{n+2}} Q_{nm}, \quad g'_{nm} + ih'_{nm} = -\frac{\mu_0}{4\pi} \frac{R^{n-1}}{n(2n+1)} q_{nm}^*, \quad (13)$$

where the primed set is considered as the “interior” Gauss coefficients appropriate with respect to the geomagnetic potential originated from the ionospheric and magnetospheric $\mathbf{J}$.

3 Gravitational Inversion in Perspective of Multipoles

The non-uniqueness of the gravitational inverse problem states the following: The external gravity field, even if completely and exactly known, cannot uniquely determine the density distribution of the body that produces the gravity field. This is an intrinsic property of a field that obeys the Laplace equation, as already treated in mathematical as well as geophysical literature (e.g., Menke 1989; Parker 1994; Backus et al. 1996).

We shall now cast this inversion problem in the multipole framework following Chao (2005). We do so in terms of the Q -multipoles, but recognize that the same argument applies also in terms of the q -multipoles on account of their symmetry on spatial

mapping. The inquiry is: By Eq. (5a), how much can one learn about the unknown $\rho(\mathbf{r})$ (of the Earth) given a complete and exact (infinite) set of multipoles (or observed Stokes coefficients)?

The multipoles are the projections of $\rho(\mathbf{r})$ onto the solid harmonics $R_{nm}(\mathbf{r})$ on a Hilbert space S that contains all possible $\rho(\mathbf{r})$ functions. However, the (infinite) set of R_{nm} that satisfy the Laplace equation does not form a complete set of basis function for S —it only spans the "harmonic" sub-space of S , leaving an orthogonal null sub-space wherefrom any element when added to any solution of $\rho(\mathbf{r})$ forms a new solution but having the same set of multipoles.

On the other hand, in lieu of the spherical-harmonic expansion Eq. (2), one can always invoke the Taylor expansion for $1/|\mathbf{r} - \mathbf{r}_0|$ (cf. Morse and Feshbach 1953; p. 1276); in the Cartesian coordinates it reads:

$$\frac{1}{|\mathbf{r} - \mathbf{r}_0|} = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \left[x_0 \frac{\partial}{\partial x} + y_0 \frac{\partial}{\partial y} + z_0 \frac{\partial}{\partial z} \right]^n \left(\frac{1}{|\mathbf{r}|} \right). \quad (14)$$

Now invoking the multinomial theorem on Eq. (14) and substituting it into Eq. (1), one gets an alternative expression for U_{ext} :

$$U_{\text{ext}}(\mathbf{r}) = G \sum_{n=0}^{\infty} \sum_{\alpha, \beta, \gamma \geq 0}^{\alpha + \beta + \gamma = n} \frac{(-1)^n}{\alpha! \beta! \gamma!} \left[\iiint_{V_0} x_0^\alpha y_0^\beta z_0^\gamma \rho(\mathbf{r}_0) dV_0 \right] \frac{\partial^n}{\partial x^\alpha \partial y^\beta \partial z^\gamma} \left(\frac{1}{|\mathbf{r}|} \right). \quad (15)$$

Equation (15) is the *moment expansion* of U_{ext} ; the quantity in the bracket is recognized as the (3-D) (polynomial) moment of order $\alpha + \beta + \gamma = n$ for the density distribution $\rho(\mathbf{r})$ in V_0 .

The multipole expansion (4) and the moment expansion (15) are of course equivalent. Equating the degree of multipole with the order of moment, we note the following: For each n , there are $2n + 1$ independent multipoles whereas $(n + 1)(n + 2)/2$ independent moments. [The latter is because the form $x^\alpha y^\beta z^\gamma$ where $\alpha + \beta + \gamma = n$ has $1 + 2 + 3 + \dots + (n + 1) = (n + 1)(n + 2)/2$ terms.] This seeming paradox can be resolved by recognizing that certain (in fact the majority of combination of) terms in (15) actually have zero contribution to the integration (1) for U (see Morse and Feshbach 1953); that is, they belong to the null sub-space of S with respect to U .

The Moment Theorem in statistics literature (e.g., Papoulis 1965, p. 214), which is closely related to the Weierstrass Approximation Theorem (e.g., Backus et al. 1996), states that a function is uniquely determined by its moments; that is, the (infinite) set of moments constitutes of projections onto a complete basis function set that spans S . In the present context that means that knowing the moments of $\rho(\mathbf{r})$ is equivalent to knowing $\rho(\mathbf{r})$ itself. Thus, in trying to determine the moments but only knowing the multipoles, the "degree of deficiency" in knowledge is $(n + 1)(n + 2)/2 - (2n + 1) = n(n - 1)/2$ -fold for each n . Conceptually, one can decompose S into disjoint sub-spaces, $S = \cup_{n=0}^{\infty} S_n$, each S_n having a null sub-space of dimension $n(n - 1)/2$ with respect to U .

For $n=0$, there is only one term in either multipole or moment expansion, i.e., the monopole=zeroth moment, corresponding to the total mass. A complete determination is assured for $n=1$ as well, whereof there are 3 dipoles and 3 first moments of $\rho(\mathbf{r})$ which correspond to the 3 components of the position vector of the center of mass. For $n=2$, there are 5 quadrupoles but 6 s moments that represent the elements of the (symmetric) inertia tensor. The multipoles are no longer sufficient to determine the corresponding moments. The degree of deficiency grows rapidly as n increases; they are summarized in Table 2. The numerability in

terms of n provides a quantitative assessment of the (gross) non-uniqueness of the 3-D gravitational inversion.

The above deficiency actually resides in the determination of the *radial* dependence of $\rho(\mathbf{r})$. That can be recognized as follows: In the special case if $\rho(\mathbf{r})$ exists only on a 2-D spherical surface, as opposed to a 3-D distribution, then it can be readily shown that Eq. (5a) can be turned around to uniquely determine the 2-D surface density distribution given the complete knowledge of the multipoles (Chao 2005). That is the assumption that makes possible the recovery of surface mass variations (in equivalent water thickness or mascons, e.g., of terrestrial water storage) from GRACE data (see Sect. 2.2), but often made implicitly or even taken for granted while lacking verification in many studies as pointed out in Chao (2014).

Earth's 3-D density models are typically inferred from tomographic models of seismic speeds by way of laboratory experimental and geochemistry evidences. Forward numerical integration of Eq. (5) can be performed on density models to yield uniquely the multipoles of course. In the case of the Q -multipoles, whether and how well the calculated multipoles agree with the observed Stokes coefficients have been an outstanding geophysical issue awaiting further investigations.

4 Multipoles Under Coordinate Transformations

Except for the exterior monopole which is a scalar, the multipoles in general are functional “components” whose values depend critically upon the chosen coordinate system. Hence they are most conveniently expressed in coordinates that take advantage of the symmetries of the body in question as indicated in Sect. 2.2; for example one would preferably stipulate the coordinate origin to coincide with the center of mass. In fact, it can be shown (Jackson 1999) that the multipole is invariant under coordinate transformation only if it is the lowest non-vanishing term. A corollary is that, given the six degrees of freedom associated with translation and rotation in 3-D (see below), one could make the exterior dipoles and tesseral quadrupoles vanish by proper coordinate transformations. But such obviously becomes impossible in separate two-body problems; for instance, a satellite's multipoles would have a rather awkward and intractable form in a planet-centric coordinates, and vice versa (cf. Fig. 2a). Therefore, in the least a coordinate translation is called for to transform the satellite's multipoles to the satellite-centric coordinates. Similarly, coordinate rotation is called for when it is desirable to align the coordinate axes with the body's principal axes.

The analytical conversion relations between two sets of spherical harmonics defined in two different coordinate systems are non-trivial; for a complete treatment we refer to those in Steinborn and Ruedenberg (1973) developed for QM. For coordinate *translation*, consider two parallel coordinate systems set apart by a translation vector $\mathbf{d}$ so that position $\mathbf{r}' = \mathbf{r} + \mathbf{d}$ (cf. Figure 4). Via the addition theorem, the expression for the translation for the regular solid harmonic R_{nm} under the given $\mathbf{d}$ is the following linear combination (e.g., Tough and Stone 1977, converted to the present normalization):

$$R_{nm}(\mathbf{r}') = (-1)^{n+m} \sum_{s=0}^n \sum_{t=-s}^s \left[\frac{4\pi(2n+1)(2n+1)!}{(2s+1)!(2n-2s+1)!} \right]^{\frac{1}{2}} \begin{pmatrix} n-s & s & n \\ m-t & t & -m \end{pmatrix} \times R_{n-s,m-t}^{\mathbf{d}} R_{st}(\mathbf{r}) \quad (16)$$

where the degree s -summation is of finite terms up to n . The information of $\mathbf{d}$ is contained in the factors $R_{n-s,m-t}^{\mathbf{d}}$, which are complex numbers (not functions) of value according to vector $\mathbf{d}$. The 6-element parenthesis Wigner $3j$ symbol (tables can be found in, e.g.,

Rotenberg et al. 1959, or obtained through online calculators, e.g., <https://www.volya.net/index.php?id=vc>) provides the selection rules that determines which $[s, t]$ harmonics get coupled in forming the original $[n, m]$ harmonic in light of symmetry, if any, in the operation of translation $\mathbf{d}$. The Wigner $3j$ symbol can be expressed in terms of the Clebsch–Gordan coefficients which appear widely in the literature (e.g., D’Urso and Adelberger, 1997; Trenkel and Speake, 1999); here we adhere to it for their common appearance in the literature of global seismology (e.g., Dahlen and Tromp 1998).

Upon performing the Q -multipole integration (Eq. 5a) to the two sides of Eq. (16), the transformation between the two sets of Stokes coefficients in the two coordinate systems set apart by translation $\mathbf{d}$ can be written as:

$$C'_{nm} + iS'_{nm} = \sum_{s=0}^n \sum_{t=-s}^s (-1)^{n-t} \left[\frac{4\pi(2-\delta_{m0})(2s+1)(2n)!}{(2-\delta_{t0})(2s)!(2n-2s+1)!} \right]^{\frac{1}{2}} \begin{pmatrix} n-s & s & n \\ m-t & t & -m \end{pmatrix} \times Y_{n-s,m-t}^{\mathbf{d}} \epsilon^{n-s} (C_{st} + iS_{st}), \quad (17)$$

where $\epsilon \equiv d/R \equiv |\mathbf{d}|/R$, and $Y_{n-s,m-t}^{\mathbf{d}}$ are a set of complex spherical-harmonic numbers according to $\mathbf{d}$. Thus, a Stokes coefficient of degree n in one coordinate system can be expressed as a linear combination of the $(n+1)^2$ Stokes coefficients with degrees $s \leq n$ from the other coordinate system. One sees in particular that the transformed Stokes coefficients depend on terms of equal or lower degree, and hence remain invariant (for example stay zero) if lower degree terms vanish in one reference frame, as mentioned above.

The summation in Eq. (17) is grouped in terms according to the power of ϵ , most convenient for discussing small translations below. For the monopole $n=0$ both sides reduce to the trivial total mass, indeed independent of the coordinate system. For the $n=1$ Q -dipoles, Eq. (17) reads:

$$C'_{1m} + iS'_{1m} = (C_{1m} + iS_{1m}) + (-1)^m \frac{\sqrt{4\pi(2-\delta_{m0})}}{3} \times \epsilon \times C_{00} \times Y_{1m}^{\mathbf{d}}, \quad m = 0, 1. \quad (18)$$

and note that $C_{00}=1$ as afore-mentioned. The only parameters of translation $\mathbf{d}$ that get involved in this coupling are its (three) degree-1 harmonics in the last term, or the center-of-mass components. A corollary is that the dipole moment is independent of coordinate system if the monopole vanishes, a circumstance true to magnetism and often to electricity when the net charge is zero.

For the Q -quadrupoles, Eq. (17) can be written in shorthand as $(C'_{2m} + iS'_{2m}) = (C_{2m} + iS_{2m}) + \epsilon \times l.c.[C_{1t} + iS_{1t}]Y_{1t}^{\mathbf{d}} + \epsilon^2 \times C_{00} \times l.c.[Y_{2t}^{\mathbf{d}}]$. Likewise, for Q -octupoles: $(C'_{3m} + iS'_{3m}) = (C_{3m} + iS_{3m}) + \epsilon \times l.c.[C_{2t} + iS_{2t}]Y_{1t}^{\mathbf{d}} + \epsilon^2 \times l.c.[C_{1t} + iS_{1t}]Y_{2t}^{\mathbf{d}} + \epsilon^3 \times C_{00} \times l.c.[Y_{3t}^{\mathbf{d}}]$; and so on for higher-degree multipoles.

The above becomes relevant to a practice that is often taken for granted in global gravity studies with respect to the Earth’s geocenter shift, the geometric departure of the center of figure relative to the center of mass. Practically only the 3 degree-1 Stokes coefficients (or the Q -dipoles) are treated for translation of coordinate origin for geocenter, as if all other coefficients were unaffected by the translation, whereas Eq. (18) actually demands they be. That can be justified as follows:

From the perspective of the infinitesimal ϵ , the *zeroth*-order ($s=n$) Eq. (17) reduces to the trivial case where the Stokes coefficient simply transform unaltered as expected. For the *second*-order effects, the couplings come from degree $n-2$ harmonics, where the only significant contribution would be from C_{00} for C'_{20} . The third-order couplings come from $n-3$ harmonics, and so on for higher-order effects in ϵ . Now given the fact

that the geocenter shifts are of the order ~ 1 cm, so ε is infinitesimal $\sim O(10^{-1})$. Compounded with the Kaula rule, all the above couplings are therefore negligibly small. That is, all those multipoles indeed remain unaltered to first order of ε ; the only exception is given in Eq. (18), i.e., the *first*-order couplings (with the backing of the monopole in the form of $C_{00} = 1$) for the Q -dipoles.

Under the coordinate translation $\mathbf{d}$, the irregular solid harmonic I_{nm} is transformed into a sum of its regular counterparts R_{st} (of infinite terms now) via the relation (Steinborn and Ruedenberg 1973; Tough and Stone 1977, converted to the present normalization):

$$I_{nm}(\mathbf{r}') = (-1)^{n+m} \sum_{s=0}^{\infty} \sum_{t=-s}^s \left[\frac{4\pi(2n+1)(2n+2s)!}{(2n)!(2s+1)!} \right]^{\frac{1}{2}} \begin{pmatrix} n+s & s & n \\ m-t & t & -m \end{pmatrix} \times I_{n+s,m-t}^{\mathbf{d}} R_{st}(\mathbf{r}) \quad (19)$$

Thus, corresponding to Eq. (17), a q -multipole in one coordinate system is transformed into an infinite linear combination of Q -multipoles in another coordinate system set apart by translation vector $\mathbf{d}$:

$$q_{nm} = \sum_{s=0}^{\infty} \sum_{t=-s}^s F_{nm}^{st} I_{n+s,m-t}^{\mathbf{d}*} Q_{st}^*, \quad (20a)$$

with the coupling coefficient:

$$F_{nm}^{st} \equiv (-1)^{n+m} \left[\frac{4\pi(2n+1)(2n+2s)!}{(2n)!(2s+1)!} \right]^{\frac{1}{2}} \begin{pmatrix} n+s & s & n \\ m-t & t & -m \end{pmatrix}. \quad (20b)$$

Equations (19, 20) apply to configurations where the field point is closer to origin than the gravitating mass element at $\mathbf{r}' = \mathbf{r} + \mathbf{d}$ (i.e., $r < r'$).

In principle a further coordinate *rotation* is entailed for association with physical quantities (i.e., Stokes coefficients) when the two bodies have distinct z directions (albeit after parallel translation), or, more generally, when the principal axes do not align. By comparison, the coordinate rotation is relatively straightforward thanks to the fact that only the *surface* harmonics of two azimuthal angles are involved (unlike the solid harmonics that also involve radial distance); the formulas are relatively well documented from QM literature (e.g., Steinborn and Ruedenberg 1973; Phinney and Burridge 1973). Specifically, the rotation operator can be expressed as a Wigner rotation matrix (e.g., Maciejewski 1995; also Boué 2017), and the transformed $[n, m']$ th surface harmonic $Y_{nm'}$ is a linear combination of surface harmonics Y_{nm} of all orders $m = -n, \dots, 0, \dots, n$ within the same degree n . The combination coefficients are inner products of $Y_{nm'}$ with Y_{nm} after acted upon by a multiplicative rotation operator expressible in rotation parameters such as the Euler angles.

In this paper we shall not deliberate on the coordinate rotation, on account of the following considerations: (i) Unlike the translation that gives rise to profoundly distinct physical situations, the coordinate rotation is only kinematic in nature in providing more precise (and more involved in appearance) descriptions than otherwise. (ii) The MICG configuration (see Sect. 5) requires no coordinate rotation as the z -axes of both bodies coincide by default. (iii) No coordinate rotation is required in tidal expansions (see Sect. 6) as the tide-generating body is approximated by a monopole (as is conventional), whereas in many planet + satellite configuration the off-alignment of the z -axes is actually quite small.

5 Two-Body: “Mantle + Inner Core”

In this section we consider a generic planet Earth model consisting of three concentric layers of mantle/outer-core/inner-core. We shall derive the two-body multipole-multipole interactions for the configuration referred to as the mantle-inner core gravitational (MICG) system (Buffett 1996; Chao 2017a)—a solid body 1 (the “inner core”) within the hollow cavity of a shell-like body 2 (the “mantle”), depicted in Fig. 3. This is a simple configuration under the body-centric coordinates with common centers of mass and a single z -axis. Besides the Earth, other studied MICG-like examples include the Saturnian moon Enceladus and Jovian satellite Europa (e.g., Van Hoolst et al. 2008), where an outer icy crust and a solid interior core are separated by a thick layer of liquid ocean.

The majority of our topics in this and following sections targets the 3-D rotational motions under torqueing. Anticipating what follows we present Table 3 at this point, which represents a unified conceptual framework. Besides the three translational degrees of freedom, a single rigid body has three degrees of freedom in rotation, a non-trivial fact true to the 3-D space (cf. Chao 1985). Thus in a rotating reference frame (such as a planet) an infinitesimal rotation manifests as three rotational normal modes (Smith 1977; Chao 1983). They are: (i) the axial spin mode along the (1-D) z -axis of spin with zero eigenfrequency, one that stays invariant under the frame rotation; and (ii) two modes in the (2-D) equatorial x - y plane arising from the coupling by the frame rotation, namely the tilt-over mode (TOM) with eigenfrequency equal (trivially) to the retrograde frame-rotation frequency itself, and the Eulerian wobble with wobble frequency depending on the moments of inertia of the body (e.g., Landau and Lifshitz 1969).

The above represent the torque-free prototype rotational modes belonging to a rigid body, to each of which correspond the rotational modes belonging to non-rigid, layered Earth or planets under additional intrinsic forcing. When subject to particular torqueing externally or internally, as is often the case of a planetary body or planetary component, the TOM-like mode’s natural period is no longer strictly retrograde diurnal in the rotating terrestrial frame, but modified to become retrograde *near*-diurnal, or kinematically of long period in the inertial frame. Referred to as “spin-over modes” by Requier et al. (2020), they manifest as *nutations* in the present context. The ratio of the two said apparent periods is ~ 400 in the case of the Earth, stemming from the oblateness (proportional to Q_{20}) of the Earth or its internal layers on the order of $\sim 1/300$. In contrast, an Eulerian wobble is in general a slow motion in the rotating frame of long period likewise modified under extra torqueing, and near-diurnal in the inertial frame, again with apparent period ratio of ~ 400 for the Earth (see, e.g., Dehant and Mathews 2015). In terms of amplitude under

Fig. 3 A generic “MICG”-type configuration: solid body 1 situated concentrically within the hollow cavity of shell-like body 2

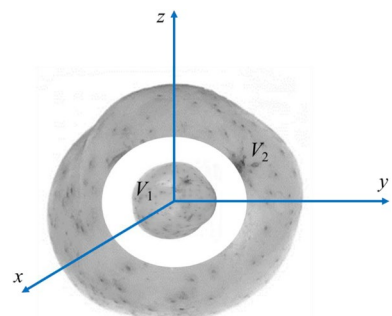


Table 3 The three prototype normal modes (left column) of free rotation belonging to a 3-D rigid body, and the corresponding cases of planetary rotations discussed in this paper (only lightly for ones in square bracket)

	MICG torque on inner core	Tidal torque on Earth	Tidal torque on Moon
Axial spin mode	Axial torsional libration ATL(5.3.1)	Spin libration (6.3.1)	Physical libration in longitude (6.3.1)
Tilt-over mode (TOM)	[unnamed] (5.3.2)	Precession/nutation (6.3.2)	Free longitudinal libration (6.3.3)
Eulerian wobble	Inner-core wobble ICW (5.3.2)	Polar libration (6.3.2)	Free latitudinal libration (6.3.3)
			Free lunar wobble (6.3.3)

The second and last columns refer to *free* rotational modes (excitation sources unknown presently); the third and fourth columns refer to *forced* rotations (by well-known tidal sources). (5.3.1), etc., indicate the section of this paper

torque-free and the conservation of angular momentum, a nutation is a rotational motion predominantly in the inertial frame, with small accompanying amplitude in the rotating frame, whereas a wobble is the opposite—a rotational motion predominantly in the rotating frame, with small accompanying amplitude in the inertial frame. The “predominant” to “small” amplitude ratio for the co-existence of the nutational and wobble components is again ~ 400 in the case of the Earth, as can be manifested by the “space cone” versus “body cone” in the Poincot’s construction (e.g., Thornton and Marion 2004; Smith 1977).

In the perspective of Table 3, the second column refers to *free-oscillation* normal modes (under extra MICG torque), presumably excited by internal geodynamo actions in the case of the Earth although the exact mechanisms are hitherto unclear. So are the last column for the free physical librations of a (generic) Moon. The excited amplitude and phase of a normal mode depend on the excitation source via a convolution process (cf. Chao 2017c); the amplitude decays with time by itself according to its characteristic Q -value until the next excitation. They are “unpredictable” in practice when lacking precise knowledge of the source. By contrast, the third and fourth columns refer to *forced* motions, in the present case by external tides with specific, known amplitudes and phases according to the orbital configurations. When a forcing’s period happens to be close to that of a certain normal mode, (near-)resonance occurs.

Back to the MICG system, the following statements should be made at the outset:

(i) While the MICG motions are actually executed by both mantle and inner core in opposite sense under the conservation of linear and angular momenta, it is practical (at least for the Earth) to speak only of the inner-core motions. That is because the inner core takes up the by-far vast share of the motions since it has much smaller mass and moment of inertia than the mantle—only $\sim 1/60$ for mass and $\sim 1/1200$ for moments of inertia for the Earth. In cases where this is not quite true, then a corresponding modification should be implemented onto the equations of motion (e.g., Van Hoolst et al. 2008), but the physics stays the same.

(ii) The MICG forcings to be derived pertain generically to the mantle+inner core per se. Nevertheless, the presence of the fluid outer core that fills the void between the two bodies ought to be taken into account, as its influences due mainly to the hydrostatic buoyancy, even in a passive manner, are profound and grave. The Archimedean buoyancy dictates that, in a fluid of density ρ_f , the net gravity felt by a body of density ρ is to be altered by the multiplicative factor $\gamma \equiv 1 - \rho_f/\rho$, reduced in value if $\rho > \rho_f$ and becoming negative (i.e., “floating”) if $\rho < \rho_f$. Thus, to account for the buoyancy effect of the outer-core, a factor $\gamma \equiv (\text{density contrast at the inner-core boundary})/(\text{mean density of the inner core}) \approx 0.60/12.8 = 0.047$ for the Earth according to the PREM model (Dziewonski and Anderson 1981) is to be multiplied to, hence greatly reduces, the forcings we shall derive below (e.g., Busse 1974; Chao 2017a).

5.1 MICG Gravitational Potential Energy

The general theory for the EM potential energy stored in a $1/r$ Coulomb field has long been established in multipolar forms in, e.g., atomic and molecular physics. For gravitation in Fig. 3, a mass element of body 1 bathed in U_{int} of body 2, U_2 , carries a gravitational potential energy $-\rho(\mathbf{r}_1)U_2(\mathbf{r}_1)dV_1$. It follows immediately that the gravitational potential energy of the two-body system can be expressed in the following simple and elegant multipole expansion:

$$E = - \iiint_{V_1} \rho(\mathbf{r}_1) U_2(\mathbf{r}_1) dV_1 = -4\pi G \sum_{n,m} \frac{1}{2n+1} Q_{nm}^{(1)} q_{nm}^{(2)} \quad (21)$$

by substituting Eq. (4) into the above integral. Exactly the same expression can be reached by reversing the roles of the two bodies, that is, by bathing body 2 in the U_{ext} of body 1. The dependence of E on the departure distance between the two bodies is implicitly contained in $q_{nm}^{(2)}$. Among the products of multipole pairs of the same $[n, m]$, each pair of $\pm m \neq 0$ multipoles of a given n adds up to the real-valued $2\text{Re}[Q_{nm}^{(1)} q_{nm}^{(2)}]$ (Re denotes real part) as the imaginary parts cancel on account of $Y_{n-m} = (-1)^m Y_{nm}^*$.

The minus sign is because this energy is negative relative to the potential set to zero at infinity, so that $|E|$ is the work to be done to separate the two bodies to infinity against mutual attractions. The requirement of minimum E demands that the two bodies, when left alone, assume stable equilibrium by conforming to certain configuration among all possibilities. In light of the Kaula rule, the equilibrium condition implies that the mantle and inner core are virtually: (i) coincident in their centers of mass in minimizing the dipole energy; (ii) aligned in their z -axes in minimizing the $Q_{20}^{(1)} q_{20}^{(2)}$ energy in Eq. (21); (iii) aligned in their equatorial principal axes in minimizing the $Q_{22}^{(1)} q_{22}^{(2)}$ energy (cf. Chao 2017a). When perturbed, the MICG system would revert to the equilibrium of minimum E , which manifests as a restoring force that gives rise to simple-harmonic motions, or normal modes. In the spirit of the Lagrangian mechanics, the generalized forces can be derived by taking the variational differentiation of E (e.g., Landau and Lifshitz 1969; Thornton and Marion 2004), although we shall instead derive them directly from definition in the below.

5.2 MICG Force and Inner-Core Slichter Mode

The static MICG force exerted on the inner core (body 1) by U_{int} of the mantle (body 2), or U_2 , is the integral:

$$\mathbf{F} = \iiint_{V_1} \rho(\mathbf{r}_1) \nabla U_2(\mathbf{r}_1) dV_1 = 4\pi G \sum_{n,m} \frac{1}{2n+1} q_{nm}^{(2)} \iiint_{V_1} \rho(\mathbf{r}_1) \nabla R_{nm}(\mathbf{r}_1) dV_1, \quad (22a)$$

by substituting in Eq. (4). Alternatively, by Newton's third law the same force can be written as the force exerted on the mantle (body 2) by U_{ext} of the inner core (body 1) with opposite sign:

$$\mathbf{F} = - \iiint_{V_2} \rho(\mathbf{r}_2) \nabla U_1(\mathbf{r}_2) dV_2 = -4\pi G \sum_{n,m} \frac{1}{2n+1} Q_{nm}^{(1)} \iiint_{V_2} \rho(\mathbf{r}_2) \nabla I_{nm}^*(\mathbf{r}_2) dV_2, \quad (22b)$$

by importing Eq. (3). That the two expressions (22a, 22b) are actually equivalent can be verified mathematically as outlined below. We note $\nabla R_{nm} = r^{n-1}(nY_{nm}\hat{\mathbf{r}} + \nabla_s Y_{nm})$ and $\nabla I_{nm} = r^{-(n+2)}[-(n+1)Y_{nm}\hat{\mathbf{r}} + \nabla_s Y_{nm}]$, which are closely associated with the vector spherical harmonics: $\mathbf{P}_{nm} \equiv Y_{nm}\hat{\mathbf{r}}$, $\mathbf{B}_{nm} \equiv r\nabla Y_{nm}$, $\mathbf{C}_{nm} \equiv \mathbf{r} \times \nabla Y_{nm}$ (e.g., Morse and Feshbach 1953; Backus et al. 1996). Meanwhile by their symmetry properties the operators of $\hat{\mathbf{r}}$ multiplication and ∇ differentiation obey selection rules that render spherical-harmonic expansions for $Y_{nm}\hat{\mathbf{r}}$ and ∇Y_{nm} in the form of $(\hat{\mathbf{x}} + \hat{\mathbf{y}}) \times l.c.[Y_{n-1,m-1}, Y_{n-1,m+1}, Y_{n+1,m-1}, Y_{n+1,m+1}] + \hat{\mathbf{z}} \times l.c.[Y_{n-1,m}, Y_{n+1,m}]$ in the Cartesian coordinates (Chao 1981). It can then be shown that all three Cartesian components of Eqs. (22a) and (22b), and hence they themselves, are equal; the somewhat tedious algebra makes use of the formulas for the vector spherical harmonics

(e.g., Morse and Feshbach 1953, p. 1898), and involves relabeling of the summation indices. Specifically, the z -component consists of $[n, m]$ multipole– $[n \pm 1, m]$ multipole pairs of the same order m , whereas the x - and y -forces originate from $[n, m]$ multipole– $[n \pm 1, m \pm 1]$ multipole pairs with both degree and order differing by 1. Note that if body 2 is spherically symmetric with respect to the origin, then the only non-zero term in Eq. (22a) is $n = m = 0$, whereupon the integral vanishes, as expected because the interior cavity within such a body 2 will have a uniform potential exerting zero force on body 1.

Implementing a virtual coordinate translation of $\delta \mathbf{r}$ to the inner body 1 and substitute Eq. (18; setting $\mathbf{d} = \delta \mathbf{r}$) into Eq. (21), one gets the MICG energy perturbation δE to first order of $\delta r/R_1$ stemming from the $n=1$ dipole terms, which, after some algebra, leads to:

$$\begin{aligned} \delta E &\approx -\frac{4\pi}{3} GM_1 \delta r \left[q_{10}^{(2)} Y_{10}^{\delta r} + \sqrt{2} \operatorname{Re} \left(q_{11}^{(2)} Y_{11}^{\delta r} \right) \right] \\ &= -\sqrt{\frac{4\pi}{3}} GM_1 \left[q_{10}^{(2)} \delta z - \sqrt{2} \operatorname{Re} \left(q_{11}^{(2)} \right) \delta x + \sqrt{2} \operatorname{Im} \left(q_{11}^{(2)} \right) \delta y \right] \\ &= \delta \mathbf{r} \cdot \left[\sqrt{\frac{4\pi}{3}} GM_1 \left\langle \sqrt{2} \operatorname{Re} \left(q_{11}^{(2)} \right), -\sqrt{2} \operatorname{Im} \left(q_{11}^{(2)} \right), -q_{10}^{(2)} \right\rangle \right] \end{aligned} \quad (23)$$

in the Cartesian coordinates. The last expression in Eq. (23) actually refers equivalently to $\delta \mathbf{r} \cdot \nabla E$ in the form of the first-order Taylor expansion: $E = E_0 + \delta \mathbf{r} \cdot \nabla E$, hence signifies the virtual work done by $\delta \mathbf{r}$ against the static MICG force given in Eq. (22), i.e., $\nabla E = -\mathbf{F} = (4\pi/3)^{1/2} GM_1 \langle \sqrt{2} \operatorname{Re} \left(q_{11}^{(2)} \right), -\sqrt{2} \operatorname{Im} \left(q_{11}^{(2)} \right), -q_{10}^{(2)} \rangle$, the vector in the square bracket. The unperturbed MICG system being in stable equilibrium, this $\mathbf{F}$ must vanish. Physically that is equivalent to say that the first-order energy perturbation δE must vanish itself, which by Eq. (23) means $q_{10}^{(2)} = q_{11}^{(2)} = 0$ for the mantle. Thus it is the second-order quadratic terms in δE , or the first-order perturbation of $\mathbf{F}$, that give rise to MICG restoring force for a 3-D harmonic translation of the inner core, known as the Slichter mode (Slichter 1961; Busse 1974). The Slichter mode has so far eluded detection from observations despite much effort (e.g., Rosat 2011; Ding and Chao 2015). The canonical Slichter mode exists under the restoring force provided by the buoyancy of the outer-core fluid that fills the void of the MICG system. Specifically, the degenerate 3-D Slichter mode is split into a triplet by the symmetry-breaking Coriolis force in the rotating terrestrial reference frame: an axial ($m=0$) singlet and two equatorial singlets corresponding to prograde $m = -1$ (prograde meaning in the same sense as Earth's spin) and retrograde $m = +1$, with eigenperiods on the order of several hours (e.g., Crossley et al. 1992). The multipole formalism of the Slichter mode and its eigenperiods awaits further investigations.

5.3 MICG Torque

Similar to Eq. (22a), the following integral gives the static MICG torque exerted on the inner core (body 1) by the U_{int} of the mantle (or U_2 of body 2):

$$\Gamma = \iiint_{V_1} \rho(\mathbf{r}_1) \mathbf{r}_1 \times \nabla U_2(\mathbf{r}_1) dV_1 = 4\pi G \sum_{n,m} \frac{1}{2n+1} q_{nm}^{(2)} \iiint_{V_1} \rho(\mathbf{r}_1) r_1^n \hat{\mathbf{r}}_1 \times \nabla_s Y_{nm}(\Omega_1) dV_1. \quad (24)$$

(Chao 2017a), into which we have substituted Eq. (4) and recognized the surface curl operator $\mathbf{r} \times \nabla = \hat{\mathbf{r}} \times \nabla_s$.

Now invoke the *angular momentum operator* $\mathbf{L} = -i \hat{\mathbf{r}} \times \nabla_s$, and the ladder relations $L_z Y_{nm} = m Y_{nm}$, and $L_{\pm} Y_{nm} \equiv (L_x \pm i L_y) Y_{nm} = \alpha_{nm}^{\pm} Y_{nm \pm 1}$ where $\alpha_{nm}^{\pm} = [(n \mp m)(n \pm m + 1)]^{1/2}$ as practiced in QM (e.g., Landau and Lifshitz 1965; Dahlen and Tromp 1998; Dehant and Mathews 2015). Here, the two Cartesian equatorial components are transcribed into the complex form that signify the prograde (+) and retrograde (−) sense of rotation with respect to East-longitude λ , with the convention that its motion factor $\exp(i\omega t)$ defines the positive frequency ω (see Munk and MacDonald 1960; Chao 1983). The 3-D Eq. (24) thus separates neatly into the (1-D) *axial* component:

$$\Gamma_z = i 4\pi G \sum_{n,m} \frac{1}{2n+1} m Q_{nm}^{(1)} q_{nm}^{(2)}, \quad (25)$$

and the (2-D) *equatorial* components:

$$\Gamma_{\pm} \equiv \Gamma_x \pm i \Gamma_y = i 4\pi G \sum_{n,m} \frac{1}{2n+1} \alpha_{nm}^{\pm} Q_{nm \pm 1}^{(1)} q_{nm}^{(2)}. \quad (26)$$

Equation (25), like Eq. (21), holds symmetry between $Q_{nm}^{(1)}$ and $q_{nm}^{(2)}$, implying that exactly the same expression with a sign reversal can be reached by switching the roles of body 1 with body 2. The factor m is switched to $-m$ owing to the complex conjugation, becoming the opposite torque as required by Newton’s third law. For $\Gamma_{\pm}$ in Eq. (26), the switching renders Γ_+ in one case to be identical to the Γ_- of the other case, and vice versa, which by Newton’s third law reflects the switching between the prograde and the retrograde sense of the equatorial torques on the two bodies.

Equations (25, 26) dictate that the monopoles, being isotropic and indifferent to the spatial orientation, do not convey any torque as both Γ_z and $\Gamma_{\pm}$ vanish for $n=m=0$ and $\alpha_{00}^{\pm} = 0$. The Q -dipoles also vanish in the geocentric center-of-mass coordinates. Thus, the non-trivial summations in Eqs. (25, 26) start with the quadrupoles.

For the axial torque Γ_z (Eq. 25), the zonal ($m=0$) multipoles contribute nothing, rightfully so because a zonal density anomaly carries no “handle” to receive axial torque. The non-zonal $\pm m \neq 0$ multipole-pair contribution to the axial torque adds up to the real-valued $-2 m \times \text{Im}(Q_{nm}^{(1)} q_{nm}^{(2)})$ (Im denotes imaginary part). Restricting to the $Q_{22}^{(1)} q_{22}^{(2)}$ term, arguably the largest term given the Kaula’s rule, the requirement of vanishing axial torque at equilibrium would signify $\text{Im}(Q_{22}^{(1)} q_{22}^{(2)}) \approx 0$ at equilibrium, implying Condition (iii) of Sect. 5.1. Equation (25) applies to, for example, the torque exerted onto a rigid inner core under super-rotation relative to a surrounding heterogeneous mantle.

For the equatorial torque $\Gamma_{\pm}$ (Eq. 26), the largest contribution to Γ_+ comes from the $[2, \pm 1]$ terms in the form of $Q_{20}^{(1)} q_{2-1}^{(2)} + Q_{21}^{(1)} q_{20}^{(2)}$ (that to Γ_- comes from $Q_{20}^{(1)} q_{21}^{(2)} + Q_{2-1}^{(1)} q_{20}^{(2)}$) upon the backing of the large magnitude of $[2, 0]$ quadrupoles. The zero equatorial torque condition at equilibrium then implies the vanishing of the above two sums, which signifies Condition (ii) of Sect. 5.1. When tidally exerted, this torque would cause the inner core to undergo a *differential* precession with respect to the mantle’s precession (cf. Sect. 6.3.2; see also Sect. 7), a motion corresponding to the prototype TOM (see Table 3).

5.3.1 Axial Torque: Inner-Core Axial Torsional Libration (ATL)

We now derive the infinitesimal MICG restoring torque that will be present for any non-zonal configurations, one which would give rise to simple-harmonic rotations, or rotational normal modes, of the inner core (as qualified with the statement (i) in Sect. 5). We insert into

Eq. (24) an *infinitesimal* displacement $\delta\mathbf{q} \times \mathbf{r}$ onto body 1 relative to body 2. This approach takes advantage of the fact that a 3-D *infinitesimal* rotation behaves like a vector along with the operation of cross product (cf. Landau and Lifshitz 1965) (whereas a general—and much more complicated—treatment of finite rotation would entail matrix transformations via, e.g., the Euler angles; see Smylie et al. 1984; Szeto and Xu 1997).

The first-order term of the Taylor expansion of the perturbation $\delta\mathbf{q} \times \mathbf{r}$, equivalent to the second angular gradient of E , is given by (Chao 2017a):

$$\begin{aligned}\delta\Gamma &= 4\pi G \sum_{n,m} \frac{1}{2n+1} q_{nm}^{(2)} \iiint_{V_1} \rho(\mathbf{r}_1) r_1^n \delta\mathbf{q} \times \mathbf{r}_1 \cdot \nabla [\hat{\mathbf{r}}_1 \times \nabla_s Y_{nm}(\Omega_1)] dV_1 \\ &= -\delta\mathbf{q} \cdot 4\pi G \sum_{n,m} \frac{1}{2n+1} q_{nm}^{(2)} \iiint_{V_1} \rho(\mathbf{r}_1) r_1^n \mathbf{L} \mathbf{L} y_{nm}(\Omega_1) dV_1\end{aligned}\quad (27)$$

in the Lagrangian description. Here we have used the relation $\delta\mathbf{q} \times \mathbf{r} \cdot \nabla = \delta\mathbf{q} \cdot \mathbf{r} \times \nabla = \delta\mathbf{q} \cdot \hat{\mathbf{r}} \times \nabla_s = i \delta\mathbf{q} \cdot \mathbf{L}$, and the double operation $\mathbf{L} \mathbf{L}$ is a second-rank tensor operator.

Taking the axial z -component and using the ladder relations as above, Eq. (27) reduces to:

$$\delta\Gamma_z = -\delta\mathbf{q} \cdot 4\pi G \sum_{n,m} \frac{1}{2n+1} m q_{nm}^{(2)} \iiint_{V_1} \rho(\mathbf{r}_1) r_1^{-(n+1)} \mathbf{L} Y_{nm}(\Omega_1) dV_1. \quad (28)$$

Now letting $\delta\mathbf{q} = \delta\lambda \hat{\mathbf{z}}$ be an infinitesimal axial torsion in longitude λ and using the ladder relation again, one obtains the equation governing a simple-harmonic motion:

$$\delta\Gamma_z = C_1 \frac{d^2}{dt^2} \delta\lambda = -\delta\lambda \times 4\pi G \sum_{n,m} \frac{1}{2n+1} m^2 Q_{nm}^{(1)} q_{nm}^{(2)} \equiv -\tilde{I}_z \delta\lambda, \quad (29)$$

where C_1 is the polar moment of inertia of the inner core. The same torque can also be reached by applying the azimuthal rotation operator to the q -multipole (Stirling 2017). This equation leads to the axial rotation of the inner core subject to the MICG restoring torque (Chao 2017a). The solution is the *axial torsional libration* (ATL; see Table 3 in perspective), called a libration on account of its generic similarity with the familiar libration of the Moon (see Sect. 6.3.2). Analogous to a classical torsion pendulum, the ATL has its characteristic *torsion constant* (its “spring constant”), or the torque per unit azimuthal longitude λ (in radian), by Eq. (29):

$$\tilde{I}_z = 4\pi G \sum_{n,m} \frac{1}{2n+1} m^2 Q_{nm}^{(1)} q_{nm}^{(2)}. \quad (30)$$

The corresponding natural angular frequency of ATL is:

$$\sigma_z = (\tilde{I}_z / C_1)^{-1/2}. \quad (31)$$

Numerically the contribution from each $\pm m \neq 0$ multipole pair to $\tilde{I}_z$ adds up to the real-valued $2m^2 \times \text{Re}[Q_{nm}^{(1)} q_{nm}^{(2)}]$. Likewise as in Eq. (25), the zonal multipoles ($m=0$), being axial-symmetric and lacking physically “handles” for exerting the ATL torque, make no contribution to $\tilde{I}_z$. The largest contribution to $\tilde{I}_z$ comes from the $[2, \pm 2]$ quadrupoles, which amounts to:

$$\tilde{I}_z \approx \frac{16\pi}{5} G [Q_{22}^{(1)} q_{22}^{(2)} + Q_{2-2}^{(1)} q_{2-2}^{(2)}] = \frac{32\pi}{5} G \text{Re}[Q_{22}^{(1)} q_{22}^{(2)}]. \quad (32)$$

The contributions of higher multipoles in expansion (30) can be augmented if deemed non-negligible.

Thus, for the simplest triaxial Earth that would possess the MICG ATL mode, the ATL frequency σ_z is proportional to $[(B-A)/C]^{1/2}$ of the inner core (since the equatorial ellipticities $B-A$ is proportional to Q_{22} ; see Sect. 2.2). Furthermore, as stated in Sect. 5, $\tilde{F}_z$ is to be multiplied by the outer-core fluid buoyancy factor $\gamma \approx 0.047$, i.e., reduced some ~ 20 -fold by “softening the MICG spring”, lengthening the ATL period by $\gamma^{-1/2}$ or some $\sim$ fivefold.

Equation of motion (29) is prescribed in the rotating terrestrial frame and has the same form as would in the inertial frame. This is a non-trivial consequence of the invariance of the axial spin eigen-mode with respect to the spin (cf. Chao 1983). Under the conservation of total angular momentum, the ATL angular displacement would manifest as an oscillation in the (mantle’s) length-of-day (LOD), with amplitude ~ 1200 times smaller than that of the inner core itself, at period $= 2\pi/\sigma_z$. Thus, following the postulation of Mound and Buffett (2003, 2006) and Chao (2017a) to equate ATL to the ~ 6 -year periodic variation detected in LOD, one can evaluate $\tilde{F}_z$ by Eq. (31) with $\sigma_z = 2\pi/6$ -year. Shih and Chao (2020) explored this postulation to deduce and evaluate q_{22} of the mantle and Q_{22} of the inner core, enabling quantitative inferences for, in particular, the density anomaly of the lower mantle + core-mantle boundary layer.

An uncertainty should be raised—the value of the inner core’s C_1 in Eq. (31) is believed to be augmented significantly from its nominal value $5.87 \times 10^{34} \text{ kg m}^2$ to a certain effective value up to $28.6 \times 10^{34} \text{ kg m}^2$ (Mound and Buffett 2003) when accounting for the outer-core Taylor-Proudman tangent cylinder encompassing the inner core. Prevalent in a rapidly spinning fluid under Coriolis force (e.g., Busse 2002), the internal motions executed within this axial tangent cylinder (such as ATL) tend to be independent of the location along the spin direction (here the z -direction). One ends up with a range of the ATL’s $\tilde{F}_z = C_1 \sigma_z^2 = 6.5\text{--}31.7 \times 10^{19} \text{ N m}$, or roughly $\sim 10^{20} \text{ N m}$, on the ~ 6 -year timescales relevant here. The actual effective value lies between the two endmembers depending on the extent of inner core coupling to the outer-core flow by electromagnetic forces.

5.3.2 Equatorial Torque: Inner-Core Wobble (ICW)

Correspondingly, we now keep the axial component in Eq. (27) unvarying and restrict the infinitesimal rotation vector $\delta\boldsymbol{\varphi}$ in latitude such that its axis lies only in the *equatorial* plane of the MICG system. After some algebra along with the ladder relations, we get (Chao 2017b):

$$\delta\Gamma_+ = -2\pi G \sum_{n,m} \frac{1}{2n+1} \alpha_{nm}^+ q_{nm}^{(2)} \left[\alpha_{nm+1}^- \delta\varphi^+ Q_{nm}^{(1)} + \alpha_{nm+1}^+ \delta\varphi^- Q_{nm+2}^{(1)} \right]. \quad (33)$$

where we have expressed $\delta\boldsymbol{\varphi}$ in the more convenient complex form $\delta\varphi^\pm = \delta\varphi_x \pm i\delta\varphi_y$. A parallel equation for $\delta\Gamma_-$ signifies the switching of the roles of the two bodies, hence redundant of the same content as $\delta\Gamma_+$ under Newton’s third law.

The monopole term in Eq. (33) vanishes, as $\alpha_{00}^\pm = 0$, as expected. We write out the dominant quadrupole terms (while remembering that higher-degree terms can be readily augmented for higher accuracy as desired):

$$\delta\Gamma_+ \approx -\frac{4\pi}{5} G \left[\left(3q_{20}^{(2)} Q_{20}^{(1)} + \sqrt{6} q_{22}^{(2)*} Q_{22}^{(1)*} \right) \delta\varphi^+ + \left(2q_{22}^{(2)*} Q_{20}^{(1)} + \sqrt{6} q_{20}^{(2)} Q_{22}^{(1)} \right) \delta\varphi^- \right], \quad (34)$$

where the tesseral $m=1$ quadrupoles are ignored given the stipulated (common) z -axis as said before. Physically, the MICG equatorial torque is exerted via the mutual handles embodied in both the $[2, 0]$ oblateness and $[2, 2]$ equatorial ellipticity. Equation (34) contains two distinct terms, leading to two entirely distinct rotational modes in their physical behavior, rightfully so in light of the fact that two degrees of freedom are associated with it (see Table 3). They are:

(i) The first term of Eq. (34) pertains to the $\delta\varphi^+$ component, which signifies a prograde sense of rotation as defined above. Within that term one can ignore the $[2, 2]$ quadrupole contribution in the presence of the dominant $[2, 0]$ oblateness-oblateness interaction between the inner core's $Q_{20}^{(1)}$ and mantle's $q_{20}^{(2)}$, and end up with $\delta\Gamma_+ \approx -(12\pi/5)GQ_{20}^{(1)}q_{20}^{(2)}\delta\varphi^+$, as was done in Chao (2017b).

Meanwhile, the inner core would have a free Eulerian wobble of its own just like the Eulerian wobble of the Earth, or more specifically the well-known Chandler wobble belonging to the non-rigid, oblate mantle with a prograde period (long relative to diurnal) in the rotating terrestrial frame. Now the inner core receives the extra MICG restoring torque, which is further and gravely reduced by the outer-core hydrostatic buoyancy factor γ as above. Entering these into the governing Euler–Liouville equation of motion (as expressed in the complex form in Munk and MacDonald 1960), the end solution (Chao 2017b) is a rotational mode known as the inner-core wobble (ICW), belonging to an axial-symmetric MICG configuration. The natural period of ICW is evaluated to be about 7 years prograde in the rotating terrestrial frame (e.g., Rochester et al. 2018, which corrected an unfortunate error in Chao 2017b, on the sign of the kinematic reference transformation from inertial to body-fixed frames). The same physics was derived via the eigen-system approach by Mathews et al. (1991).

Neither of the two quantities in the deciding factor $Q_{20}^{(1)}q_{20}^{(2)}$ is observed directly, however the associated $[2, 0]$ density profiles can be determined layer-by-layer to a high accuracy by virtue of rotation-hydrostatic equilibrium through Clairaut's equation (e.g., Bullen 1975; Rochester et al. 2018) based on, say, the PREM Earth density model, making the ~ 7 -year prograde period of ICW well asserted. The ICW, or any possible evidences or consequences, have yet to be detected by modern geodetic observations.

(ii) The second term of Eq. (34), $\delta\Gamma_+ \approx -\frac{4\pi}{5}G\left(2q_{22}^{(2)*}Q_{20}^{(1)} + \sqrt{6}q_{20}^{(2)}Q_{22}^{(1)}\right)\delta\varphi^-$, pertains to the $\delta\varphi^-$ component, which signifies a retrograde sense of rotation. This restoring torque, which would be absent under axial symmetry, stems from the oblateness-equatorial ellipticity interactions, hence is much weaker than that for the ICW above (and to be further reduced by γ), implying a much longer eigenperiod. The resultant motion corresponds to the prototype TOM of the inner core (see Table 3), manifesting a long-period slow nutation in the inertial frame but in any event near-diurnal retrograde in the rotating terrestrial frame as it should. The period with respect to the inertial frame depends on the relevant multipoles above (among which the $[2, 2]$ multipoles are undetermined), prograde if positive or retrograde if negative. This nutational mode is hitherto unnamed (as so entered in Table 3), and should be possessed by at least a triaxial Earth model. It is not to be confused with the free-inner core nutation, an axial-symmetric fluid pressure mode (Mathews et al. 1991) that is absent in our present context of gravitation (see Sect. 7).

6 Two-Body: “Planet + Satellite”

In this section we investigate the two-body interaction in a “planet + satellite” configuration depicted in Fig. 4, for example Earth + Moon. In the latter case the influence of the Sun (or other planets for that matter) is to be added by simple superposition, and hence is not considered explicitly.

The multipole expansions in Sect. 5 are derived for two-body interactions in general (cf. Trenkel and Speake 1999); it should be noted that no part of the derivation is restricted to the peculiarity of the MICG system per se. Therefore the formulas continue to be valid here—except, as cited in Sect. 4, that one now needs to perform the coordinate transformations so that the satellite’s multipoles awkwardly expressed in the planet-centric coordinates could be expanded into a superposition of multipoles expressed in the “proper” satellite-centric coordinates.

6.1 Gravitational Potential and Energy

Performing a coordinate translation $\mathbf{d}$, the position vector of the satellite in the planet-centric coordinates, by Eq. (20) onto Eq. (21) for the general two-body gravitational potential energy, one gets:

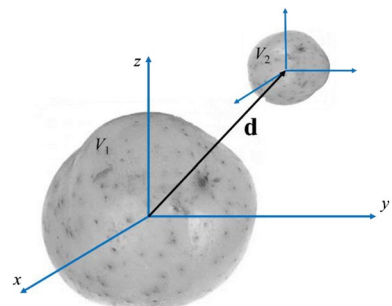
$$E = -4\pi G \sum_{nm} \frac{1}{2n+1} Q_{nm}^{(1)} q_{nm}^{(2)} = -4\pi G \sum_{nm} \sum_{s,t} \frac{1}{2n+1} F_{nm}^{st} I_{n+s,m-t}^{d*} Q_{nm}^{(1)} Q_{st}^{(2)*} \quad (35)$$

Each term of Eq. (35) represents neatly the interaction between the planet’s Q -multipole and satellite’s Q -multipole all expressed in the respective body-centric coordinates.

Now we target the leading term pertaining to the two original monopoles (in the middle expression of Eq. 35), i.e., the sole term when we let body 1 (with $Q_{00}^{(1)}$) to be a point “test mass” of mass M_1 . Then upon dividing off M_1 and reversing the roles of the two bodies, the right side of Eq. (35) reduces to the gravitational potential produced by (an arbitrary) body 2 of mass M_2 , where $\mathbf{d}$ now signifies the location of the field point $\mathbf{r}$. When written out alternatively in Cartesian tensor form, that leads to the following formula that is familiar in EM (e.g., Hirschfelder et al. 1954; Jackson 1999; Snieder 2001):

$$U(\mathbf{r}) \approx -4\pi G Q_{00}^{(1)} q_{00}^{(2)} / M_1 = -GM_2/d + G(\mathbf{p}_2 \cdot \hat{\mathbf{d}})/d^2 - \frac{1}{2} G \sum_{i,j=1}^3 \mathbf{p}_{ij}^{(2)} \hat{d}_i \hat{d}_j / d^3 + \dots \quad (36)$$

Fig. 4 Two separate bodies of a planet (V_1 of body 1) in the planet-centric coordinates + satellite (V_2 of body 2) in the satellite-centric coordinates, set apart by the planet-satellite displacement vector $\mathbf{d}$



The first term on the right side is the well-known monopole potential corresponding to monopole-monopole energy. The second term, where $\mathbf{p}$ is the dipole moment tensor, corresponds to the monopole-dipole energy appearing widely in EM. The third term corresponding to the monopole-quadrupole energy is actually recognized to be equivalent to the well-known MacCullagh's formula (e.g., Stacey and Davis 2008; Dehant and Mathews 2015) in gravitation, here in terms of the traceless quadrupole moment tensor $\mathbf{P}$. We proclaim here that Eq. (36) will reappear in disguise later as Eq. (38) pertaining to the tidal potential.

Moving on to quadrupoles, it can be shown that, by way of the next term $-4\pi G Q_{20} q_{20}/5$, one has the zonal-quadrupole-monopole energy $= (G/2d^3)[C - (A + B)/2]_1 M_2 (3\cos^2\theta_d - 1)$, the zonal-quadrupole-zonal-quadrupole energy $= -(3G/4d^5)[C - (A + B)/2]_1 [C - (A + B)/2]_2 (35\cos^4\theta_d - 30\cos^2\theta_d + 3)$, the zonal-quadrupole-sectoral-quadrupole energy $= (15G/8d^5)[C - (A + B)/2]_1 (B - A)_2 \sin^2\theta_d (7\cos^2\theta_d - 1)\cos 2\lambda_d$, etc. As stipulated, the Q -dipole terms in Eq. (35) vanish upon the selection of the center-of-mass coordinates; otherwise all the terms in Eq. (35) would have respective "shifted" values (except the monopoles) but their sum being a scalar remains the same of course.

Now we treat the *self-energy* of a "charge" that produces a conservative $1/r$ field (e.g., Feynman et al. 1964), in the case of gravitation due to a *single* body given by:

$$E_s = -\frac{1}{2} \int_V \rho U dV. \quad (37)$$

Here U is the gravitational potential generated by all mass particles of the body other than the one in consideration, and the factor $1/2$ is to undo the double count of the particles. E_s is a negative energy whose absolute value is the work to be done to disassemble all the mass particles gravitationally and send them out to infinity against the mutual gravitational attraction, or equivalently the energy liberated when all the mass particles were accreted from infinity into the condensed configuration.

A uniform sphere has the well-known $E_s = -3/5 GM^2/R$, obtained by integrating the work done to disassemble shell by spherical shell, which amounts to -2.24×10^{32} J for the uniform Earth, or 2.5×10^{15} kg in (relativistic) equivalent mass loss over the life of Earth. Further gravitational energy was liberated by the material differentiation process during the course of the Earth history, rendering the present-day $E_s \sim 10\%$ more negative than the above (Marchenko and Zayats 2011).

Following the general forms for EM Coulomb self-energy by Carlson and Morley (1963; see also Rubincam 1979), E_s can be partitioned into individual components symbolically expressed as $E_s = E_{\text{monopole}} + E_{\text{dipoles}} + E_{\text{quadrupoles}} + E_{\text{octupoles}} + E_{\text{higher-multipoles}} \dots$ thanks to the orthogonality of spherical harmonics, whereof each term involves a double integral of ρ over the same volume in contrast to two separate integrals over the two bodies. Chao (2014) obtained the zonal-quadrupole component of E_s , the largest term next to E_{monopole} for the oblate spheroidal Earth: $E_{\text{oblateness}} = 4/45 \times (\text{geometrical oblateness})^2 |E_{\text{monopole}}| \approx 10^{-6} |E_{\text{monopole}}| = +2.5 \times 10^{26}$ J. This is the (positive) energy at disposal as the Earth's J_2 decreases in magnitude, whose observed rate $-2.8 \times 10^{-11} \text{ year}^{-1}$ due mainly to the glacial isostatic adjustment process dissipates $E_{\text{oblateness}}$ at a rate of -200 GW (Chao 2014). Within the budget of the mantle energetics that amounts to as much as $\sim 20\%$ of the total convective power required to drive the entire plate tectonics (estimated to be of ~ 1 TW; e.g., Stacey and Davis 2008).

6.2 Tidal Potential and Tidal Force

Now we derive the general multipole formulation of the tidal potential and the various tidal phenomena, but largely only to the extent of performing the planet-satellite coordinate translation $\mathbf{d}$ (see Sect. 4 and Fig. 4). We shall not delve into the subsequent development of the tide into tidal constituents, which consists of decomposing the time-variable $\mathbf{d}$ in terms of the time-variable celestial ephemerides parameters along with Earth's diurnal rotation whereof textbook literature abounds (e.g., Melchior 1983; Dehant and Mathews 2015).

The tidal field arises from imposing the gravitational potential generated by the satellite (body 2) onto the volume occupied by the planet (body 1). Substituting the coordinate translation Eq. (20) into Eq. (4), originally in terms of the satellite's q -multipoles in the planet-centric coordinates, yields the general expression for the tide generating potential:

$$U(\mathbf{r}) = 4\pi G \sum_{n,m} \frac{r^n}{2n+1} q_{nm}^{(2)} Y_{nm}(\Omega) = 4\pi G \sum_{n,m} \sum_{s,t} \frac{r^n}{2n+1} F_{nm}^{st} I_{n+s,m-t}^{\mathbf{d}*} Q_{st}^{(2)*} Y_{nm}(\Omega), \quad (38)$$

where the satellite's $Q_{st}^{(2)}$ multipoles are now expressed properly in the satellite-centric coordinates. The right-side sum over n,m pertains to the tide-receiving volume of body 1, while that over s,t pertains to the tide-generating body 2; they are connected term-by-term via vector $\mathbf{d}$ in the form of the factor $I_{n+s,m-t}^{\mathbf{d}*}$.

The conventional “standard” formulation of the luni-solar tides on Earth approximates the Moon (or Sun) as a spherically-symmetric body or equivalently a point mass (hence no further coordinate rotation is needed). That means retaining only the Moon's Q -monopole (with $s=t=0$) in Eq. (38), i.e., $Q_{00}^{(2)} = \sqrt{1/4\pi} M_2$, for which the 3- j symbol in Eq. (20b) is $\begin{pmatrix} n & 0 & n \\ m & 0 & -m \end{pmatrix} = (-1)^{n-m} \frac{1}{\sqrt{2n+1}}$. Equation (38) in this approximation reduces to the sum:

$$\begin{aligned} U(\mathbf{r}) &= \frac{GM_2}{d} \sum_{nm} \frac{4\pi}{2n+1} \left(\frac{r}{d}\right)^n Y_{nm}^*(\Omega_d) Y_{nm}(\Omega) \\ &= \frac{GM_2}{d} \left[1 + \frac{4\pi}{3} \left(\frac{r}{d}\right) (Y_{10}^{d*} Y_{10} + 2\text{Re}(Y_{11}^{d*} Y_{11})) \right. \\ &\quad \left. + \frac{4\pi}{5} \left(\frac{r}{d}\right)^2 (Y_{20}^{d*} Y_{20} + 2\text{Re}(Y_{21}^{d*} Y_{21}) + 2\text{Re}(Y_{22}^{d*} Y_{22})) \right] + \dots \end{aligned} \quad (39)$$

where $d \equiv |\mathbf{d}|$ is the planet-satellite distance. The $Y^{\mathbf{d}}$'s in the present context refer to the direction of the (moving) sub-lunar (or sub-solar) point on the Earth, itself rotating.

Equation (39) is indeed a conventional formula for the tide generating potential (e.g., Efroimsky and Williams 2009), but actually in disguise of Eq. (36) as said above. The first term in Eq. (39) is a constant bearing no force (with vanishing gradient). The second term gives rise to the central force providing for the Keplerian orbital motion. The third term, proportional to d^{-3} , is the leading term of the tidal potential. The deciding factor for the tidal amplitude is thus M_2/d^3 ; that is why the lunar tide is 2.2 times stronger than the solar tide on Earth. Already put in the planet-centric coordinates this tidal potential is conducive for further development into tidal constituents.

The above monopole approximation is sufficiently accurate when d much exceeds the size of the bodies, such as in the Earth–Moon system where $d \approx 60R$. For higher accuracy, one can work out from Eq. (39) the quadrupole terms proportional to d^{-4} , which for Earth amounts to two orders of magnitude smaller than the leading tide, etc.

Alternatively, by setting the z -axis along $\mathbf{d}$ (so $\theta_d = 0$ and non-zonal Y^d vanish), the tidal potential reduces to a familiar form:

$$U_{\text{tide}}(\mathbf{r}) = GM_2 \left(\frac{r^2}{d^3} \right) P_2(\cos \theta), \quad (40)$$

where the Legendre function $P_2 = P_{20}$, and θ is to be understood as the angle subtended by the field point from the planet-satellite axis. Equation (40) is alternatively often taken in textbooks as the starting point for ensuing development of the tidal constituents (e.g., Melchior 1983; Murray and Dermott 1999; Stacey and Davis 2008).

The next set of summation upon returning to Eq. (38) for $s=2$ are to be invoked in evaluating the quadrupolar effects (such as oblateness) of the tide-generating source, and so on for higher s due to additional finer off-sphericity.

Equation (38) applies symmetrically to the reverse situation, i.e., to describe the tide potential generated by the planet over the volume of the satellite, just by swapping the nametags. A “trivial” case is when the satellite is reduced to a point mass, and the problem reduces to that of the classical satellite orbit in the Earth’s gravitational field Eq. (3). As is familiar and considered as tidal effects in the general sense of the term, the Earth’s oblateness then gives rise to the precessions of node and periaapsis in the Moon’s orbit, or more prominently in the orbits of artificial satellites that are closer to the Earth. Going to the quadrupolar terms, an interesting case is the tide generated by the highly oblate planet Saturn onto its near moons. The effect can reach as high as a fraction of 1% of the main monopolar tide; see Sect. 6.3.1.

6.3 Tidal Torques

By the same token, Eq. (24) holds for two-body tidal torques in general; so do its derivatives Eq. (25) for the axial component and Eq. (26) for the equatorial components, as well as the corresponding Eq. (27) for the infinitesimal restoring torque and its components (28) and (33). The extra procedure entailed here is to carry out the planet-to-satellite coordinate translation as above.

6.3.1 Axial Torque

Upon substituting in Eq. (20) for coordinate translation $\mathbf{d}$, Eq. (25) leads to the axial tidal torque:

$$\begin{aligned} \Gamma_z &= i 4\pi G \sum_{n,m} \frac{1}{2n+1} m Q_{nm}^{(1)} q_{nm}^{(2)} \\ &\approx -\frac{16\pi G}{5} \text{Im} \left[Q_{22}^{(1)} q_{22}^{(2)} \right] = -\frac{16\pi G}{5} \text{Im} \left[Q_{22}^{(1)} \sum_{s,t} F_{22}^{st} I_{2+s,2-t}^{d*} Q_{st}^{(2)*} \right] \\ &\approx -\frac{16\pi G}{5} \text{Im} \left[F_{22}^{00} I_{22}^{d*} Q_{22}^{(1)} Q_{00}^{(2)*} \right] = \frac{3 G M_2 (B-A)_1}{2 d^3} \sin^2 \theta_d \sin 2(\lambda_d - \Lambda_1). \end{aligned} \quad (41)$$

where θ_d and λ_d are, respectively, the co-latitude and longitude of the sub-lunar point on Earth, and Λ_1 is the longitude of the Earth’s A axis. In deriving Eq. (41) we have made two approximations ($\approx$) by retaining only the leading terms and ignoring the rest: (i) We retain the largest non-zonal quadrupole $Q_{22}^{(1)}$ for the tide-receiving body 1 (the zonal multipoles, $m=0$, have no handle to receive the axial torque), in effect taking it for a triaxial body.

There are terms due to the tesseral quadrupole $Q_{21}^{(1)}$ that vanish upon the choice of the z -axis to align with the polar principal (C) axis. (ii) We retain the dominant monopole $Q_{00}^{(2)}$ ($s=t=0$) for the tide-generating body 2, just as done in Sect. 6.2 for the conventional tidal potential. We have also substitute in the physical parameters for $Q_{22}^{(1)}$ and $Q_{00}^{(2)}$ given in Sect. 2.2.

The last expression of Eq. (41) duly reflects the tidal origin by the distance dependence of d^{-3} (see Sect. 6.2), and its semidiurnal nature by the angular dependence of $\sin 2(\lambda_d - \Lambda_1)$ on the phase angle between the A axis and the Earth–Moon line. The axial torque is exerted on the equatorial ellipticity $B - A$, and assumes maximum at equatorial crossing $\theta_d = 90^\circ$ by the factor $\sin^2 \theta_d$. Below we inspect two classical cases of planetary rotational motions subject to this axial tidal torque (cf. Table 3):

(i) Earth’s spin libration (Chao et al. 1991): The same formula above, $\tau_z = \frac{3GM_M}{2d^3} (B - A)_E \sin^2 \theta_d \sin 2(\lambda_d - \Lambda_E)$, has been derived through a conventional approach and employed by Chao et al. (1991) to evaluate the Earth’s (suffix E) tidal spin libration, the forced spin variation by external luni-solar (suffix M) tidal torque (distinct from the MICG ATL due to internal processes that conserve angular momentum). Simplistically, the factor $\sin 2(\lambda_d - \Lambda_1)$ states that the lunar tidal torque on Earth’s $B - A$ under diurnal spin will accelerate the Earth’s spin when its A axis approaches the Earth–Moon line, and decelerate the spin after the A axis passes that line.

The result is a semidiurnal Earth libration in the longitude superimposed upon the otherwise uniform diurnal spin. The amplitude is in proportion to the triaxiality parameter $(B - A)/C$ of the Earth—or that of the mantle-only if the cores do not participate in the motion at the timescale of concern; the difference would be on the order of a few percent. The (integrated) amplitude in terms of Universal Time or angular displacement (in proportion to the spin period squared) is about 0.09 milliarcsecond (mas; 1 mas corresponds to ~ 3 cm on Earth surface) peak-to-peak, the largest tidal constituent of which is the lunar semidiurnal M2, followed by the solar semidiurnal S2 (Chao et al. 1991). It is interesting to note that this angular displacement for planet Venus (with respect to solar tide) is as much as 4×10^4 times larger because of Venus’ long spin period.

The tidal spin libration contributes to the tidal energy dissipation in the mantle due to its slight anelasticity. The Earth’s (including the fluid ocean) response to the luni-solar tidal potential is not instantaneous but rather accompanied by rotational energy dissipation manifesting as slight phase retardation. The Earth’s spin slows down and LOD lengthens secularly over geological time as a result.

(ii) Forced physical libration in longitude of the Moon: Superposed onto its classical Cassini state, the Moon executes physical librations (distinct from the optical or geometrical libration that is perhaps more commonly-known) in both longitude and latitude (see below) under the tidal torques exerted by the Earth (and Sun and other planets). The complete development of the physical librations into tidal constituents in the lunar geographical coordinates can be found in Eckhardt (1981; using the Euler equation) and Moons (1982; using the Hamiltonian approach), notwithstanding some discrepancies between them. Rambaux and Williams (2011) have reported the detection of these physical librations (as well as the free physical librations—see below) from the lunar laser ranging (LLR; see Dickey et al. 1994) observations made since the astronauts of the Apollo Project (completed in 1972) deployed the retroreflectors on the face of Moon.

As before, the same expression (41) holds equally valid to its reverse situation upon switching the roles of the planet and satellite, giving the physical libration in longitude of the triaxial Moon driven by the tidal torque of the (spherical or point-mass) Earth, or for that matter a synchronous-rotating moon relative to the primary planet. Specifically, the

factor $\sin 2(\lambda_d - \Lambda_1)$ demands that $\lambda_d = \Lambda_1$, i.e., the Moon's A axis points to the Earth–Moon line, while at equilibrium with vanishing τ_z . Setting $\Lambda_1 = 0^\circ$ at the mean sub-Earth point on the Moon, this torque drives the Moon's libration of longitude of amplitude $\frac{3GM_E}{2d^3} \left(\frac{B-A}{C} \right)_M \sin^2 \theta_d \sin 2\lambda_d$ (e.g., Murray and Dermott 1999), to be further decomposed into tidal constituents (Eckhardt 1981). The LLR observations of the latter thus leads to the Moon's triaxiality parameter $(B-A)/C = 227.88 \times 10^{-6}$ (Dickey et al. 1994; Bills 1995), about 10 times of the Earth's 21.9×10^{-6} (Liu and Chao 1991).

Although of academic interest, one can throw in the $s=2$ quadrupole terms in Eq. (41) if one is to consider a triaxial tide-generating planet. Letting $\theta_d \approx 90^\circ$ (i.e., small spin–orbit obliquity), doing that leads to $\Gamma_z \approx (3GM_E/2d^3) (B-A)_M [(1+\mu_1) \sin 2\lambda_E + \mu_2 \sin 4\lambda_E]$, with the small (< 1) factor $\mu_1 = (5/2)J_{2(E)}(R_E/d)^2$, and $\mu_2 = (35/2)C_{22(E)}(R_E/d)^2$ giving the higher-order effects. For example, to the highly oblate Saturn ($J_2 \approx 0.0163$) and its relatively near moons Mimas ($d \approx 3.18 R$) and Enceladus ($d \approx 4.09 R$), μ_1 amounts to about 0.43% and 0.26%, respectively, compared to the corresponding $\sim 7.5 \times 10^{-7}$ for Earth's Moon.

6.3.2 Equatorial Torque

Similar to Eq. (41), substituting Eq. (20) into (26), one arrives at:

$$\begin{aligned} \Gamma_+ \equiv \Gamma_x \pm i\Gamma_y &= i4\pi G \sum_{n,m} \frac{1}{2n+1} \alpha_{nm}^\pm Q_{nm\pm 1}^{(1)} q_{nm}^{(2)} \\ &\approx -\frac{i4\pi G}{5} \left[\sqrt{6} Q_{20}^{(1)} q_{21}^{(2)*} - 2Q_{22}^{(1)} q_{21}^{(2)} \right] \\ &= -\frac{i4\pi G}{5} \left[\sqrt{6} Q_{20}^{(1)} \sum_{s,t} F_{21}^{st} I_{2+s,1-t}^d Q_{st}^{(2)} - 2Q_{22}^{(1)} \sum_{s,t} F_{21}^{st} I_{2+s,1-t}^{d*} Q_{st}^{(2)*} \right] \\ &\approx -\frac{i4\pi G}{5} \left[\sqrt{6} F_{21}^{00} I_{21}^d Q_{20}^{(1)} Q_{00}^{(2)} - 2F_{21}^{00} I_{21}^{d*} Q_{22}^{(1)} Q_{00}^{(2)*} \right] \\ &= -\frac{3iGM_2}{4d^3} \sin 2\theta_d \left[2 \exp(i\lambda_d) \left(C - \frac{A+B}{2} \right)_1 + \exp(-i\lambda_d) (B-A)_1 \exp(i2\Lambda_1) \right]. \end{aligned} \quad (42)$$

Here we have again taken the approximations ($\approx$) of a triaxial Earth whose $[2, 0]$ and $[2, 2]$ quadrupoles receive the tidal torque, and a monopolar tide-generating Moon (or Sun) (the $[2, 1]$ tesseral vanishes upon the choice of the z -axis as above). Again, the distance dependence of d^{-3} attests to its tidal identity. The factor $\sin 2\theta_d$ dictates that this equatorial torque exists ($\sin 2\theta_d \neq 0$) when and only when there exists a non-zero obliquity (i.e., $\theta_d \neq 90^\circ$) of the lunar (or solar) orbit relative to the Earth's equatorial plane, which reaches maximum at the solstices. Again, the last equality of Eq. (42) has been derived by Chao et al. (1991; their Eq. 4b in terms of un-normalized Stokes coefficients) following the conventional approach.

Equation (42) contains two terms that are entirely distinct in their dependence on λ_d subject to the equatorial torque, hence distinct in their physical behavior. Below we inspect the two classical cases of planetary rotations corresponding in the present context to the two equatorial degrees of freedom for rotation; see Table 3. They are:

(i) Astronomical precession/nutation: The first term in the last expression of Eq. (42), i.e.,

$$\Gamma_+ \approx -\frac{3iGM_M}{2d^3} \sin 2\theta_d \exp(i\lambda_d) \left(C - \frac{A+B}{2}\right)_E \quad (43)$$

gives the torque exerted by the monopolar Moon (or Sun) M_M , upon the Earth's oblateness (or “equatorial bulge” as often described) embodied in the quadrupolar $[C - (A+B)/2] \equiv J_2 MR^2$. This expression is equivalent to, e.g., Eq. (2.39) of Melchior (1983) but in a more compact complex form accentuating its rotational motion.

Tidal torque (43) gives rise to the astronomical precession and nutations of the Earth. In the perspective of Table 3, the precession/nutation corresponds to the prototype TOM with retrograde near-diurnal periods in the rotating terrestrial frame, as is described by the factor $\exp(i\lambda_d)$ in the present convention that defines the positive frequency. In the inertia space it signifies a near-zero frequency, or nearly infinite period, for the Sun's longitude. Letting the y-axis be the axis pointing to the equinox of the Sun's orbit around Earth (and x-axis to the solstice), Stacey and Davis (2008) has shown that, letting $A=B$ for simplicity, $\Gamma_y \approx -\frac{3GM_M}{4d^3} \sin 2\theta_d (C-A)_E$ when averaged over long term. Inputting it into the Euler equation of motion one gets the astronomical *precession* of equinox, in complete analogy to a precessing top. The rate of precession due to the Sun is approximately $-2/3 * [(\text{Sun's orbital rate})^2 / (\text{Earth's spin rate})] * [\text{Earth's } (C-A)/C] * \cos(\text{Earth obliquity})$. The Moon contributes in a similar way, but more than twice as strong. The Earth has the observed retrograde period of 25,772 years for the precession of equinox, leading to the determination of its dynamic oblateness $(C-A)/C = 1/305.51$.

The torque Γ_x on the other hand introduces “nodding” of the Earth in accordance with the luni-solar celestial fluctuations up to the longest tidal period of 18.6 years, constituting the astronomical *nutations* that are superposed onto the mean precessional path. The factor $\sin 2\theta_d$ dictates that the torque switches off at every equatorial crossing of the Sun or Moon, hence the largest nutation terms are of the semi-periods of the principal tidal periods.

The reverse tide, i.e., the equatorial tidal torque exerted on the Moon's oblateness by the Earth or Sun can be formulated and understood similarly. That results in the Moon's retrograde precession at period of 18.6 years in the inertial frame, constituting one of the Cassini's laws of the Moon's motion (e.g., Eckhardt 1981).

(ii) Polar libration in Earth's polar motion (Chao et al. 1991): The second term in the last expression of Eq. (42), i.e.,

$$\Gamma_+ \approx -\frac{3iGM_M}{4d^3} \sin 2\theta_d \exp(-i\lambda_d) (B-A)_E \exp(i2\Lambda_E) \quad (44)$$

is, as should, identical to that derived by Chao et al. (1991, by conventional approach; see also Kinoshita 1977, using Hamiltonian approach). This tidal torque is exerted upon the Earth's equatorial ellipticity $B-A$ by a monopolar Moon (or Sun) of M_M . The factor $\exp(-i\lambda_d)$ signifies a diurnal, nominally prograde propagation in the rotating frame opposite to $\exp(i\lambda_d)$ and has no secular effect as does the precession. Implemented into the Euler-Liouville equation (Munk and MacDonald 1960; cf. Sect. 5.3.2), this external torque would drive the Earth's wobble or polar motion in the context of Table 3. A transfer function (in the sense of generalized Love number) of 0.480 is to be multiplied to the torque to account for the elastic feedback effect of the Earth (Chao et al. 1991). The end result is the polar libration, in proportion to $(B-A)/A$, of peak-to-peak amplitude about 0.06 mas, the largest tidal constituent of which is the luni-solar diurnal K1, followed by the lunar diurnal O1. Note that the tidal-driven polar librations that fall in the retrograde diurnal band in the

rotating frame are actual polar motions by definition; they are not nutations except that they are observationally inseparable from nutations.

The reverse tide, i.e., the Moon's polar physical libration in latitude driven by the Earth's equatorial tidal torque, can be determined in exactly the same manner, which shall not be dwelt upon here but entered in Table 3. A complete tidal decomposition can be found in Eckhardt (1981).

6.3.3 Free Physical Librations of the Moon

The tidal torques above are responsible for the forced motions. To the tidally “locked” synchronous-rotating moons, they also provide the restoring torques to execute simple-harmonic oscillations about the stable equilibrium orientation with certain natural periods behaving like a physical pendulum. That constitutes the *free* physical libration modes, which have largely been disregarded (Moons 1982). Three such modes are expected corresponding to the three rotational degrees of freedom (cf. Table 3); they have been detected in the LLR data as reported by Rambaux and Williams (2011). It should be emphasized that, as of any normal mode, the observed motion (i.e., non-zero amplitude) of these modes owes the existence to some actual excitation sources, identified or not as the case may be, and will dissipate with time according to the characteristic Q -values. Below we inspect the free physical librations of the Moon:

(i) Free libration of longitude: This motion is the planet+satellite counterpart of the MICG ATL (Sect. 5.3.1). Thus, Eq. (30), upon implementing the coordinate translation $\mathbf{d}$ (Eq. 20) as before and switching the mutual roles of Earth (suffix E) and Moon (suffix M), yields the associated axial torsion constant:

$$\begin{aligned}\tilde{I}_z &= 4\pi G \sum_{n,m} \frac{1}{2n+1} m^2 Q_{nm}^{(M)} q_{nm}^{(E)} \\ &= 4\pi G \sum_{n,m} \sum_{s,t} \frac{m^2}{2n+1} F_{nm}^{st} I_{n+s,m-t}^{d*} Q_{nm}^{(M)} Q_{st}^{(E)*} \approx \frac{3GM_E(B-A)_M}{d^3}\end{aligned}\quad (45)$$

Notice the formal similarity to Eq. (35) for energy and Eq. (41) for static tidal torque saving the multiplicative factors; hence the qualitative discussions there apply here as well. Specifically, in the last approximation ($\approx$) we have taken the tide-generating Earth as a monopole and the Moon as a quadrupolar triaxial body with equatorial ellipticity $(B-A)$, and let $\theta_d = 90^\circ$ on average and $\Lambda_M = 0^\circ$ for the synchronous-rotating Moon.

Therefore, similar to Eq. (31) for ATL, we have $\sigma_z^2 = \tilde{I}_z/C_M = \frac{3GM_E}{d^3} \left(\frac{B-A}{C}\right)_M$ for the square of the natural angular frequency of the Moon's free longitudinal libration, or $\sigma_z = [3(B-A)/C]_M^{1/2} n$, where $n = [GM_E/d^3]^{1/2}$ is the Keplerian mean orbital motion of the Moon (at 27.3-day period). That corresponds to a period of 2.86 years according to the value of $[(B-A)/C]_M$ given in Dickey et al. (1994). Matching well, Rambaux and Williams (2011) reported the detection of this motion at the period of 2.89 years in the LLR data, with amplitude above 1 arcsecond. It is debatable whether this dynamics-related period pertains to the whole Moon or its mantle only (assuming the presence of a fluid core) at the relevant timescale. Gravity unfortunately is unable to offer a resolution since it alone does not determine the global C , as it does the global $B-A$.

(ii) Eulerian wobble: Similarly, by switching the roles of the two bodies, Eqs. (33, 34) represent the tidal torque by the Earth to give rise to the Moon's free libration in the

equatorial plan about its locked Cassini orientation. Again, there are two degrees of freedom. The first term of Eq. (34) pertaining to the $\delta\varphi^+$ component becomes:

$$\delta\Gamma_+ \approx -\frac{4\pi G}{5} \left[\delta\varphi^+ \left(\sqrt{6} q_{22}^{(E)*} Q_{22}^{(M)*} + 3q_{20}^{(E)} Q_{20}^{(M)} \right) \right], \quad (46)$$

where we have made the approximations ($\approx$) for a quadrupolar triaxial Moon and a monopolar tide-generating Earth, and dropped the tesseral quadrupoles as before. Upon implementing coordinate translation (20), this torque gives rise to the Moon’s Eulerian wobble, as the counterpart of ICW in Sect. 5.3.2. We shall omit the detail derivations which are outlined in Sect. 5.3.2, but state that the resultant natural angular frequency (subject to locking) is twice of would be a free wobble (Rambaux and Williams 2011; cf. Chao 1983), i.e., $2[(C-A)(C-B)/AB]_M^{1/2} n$, which corresponds to a slow motion at the prograde period of 75 years in the 27.3-day lunar rotating frame. The Rambaux and Williams’ (2011) LLR-observation put an amplitude over 8 arcseconds, source(s) unknown.

(iii) Free libration of latitude: By the same token, the second term of Eq. (34) pertaining to the $\delta\varphi^-$ component:

$$\delta\Gamma_+ \approx -\frac{8\pi}{5} G \left(q_{22}^{(2)*} Q_{20}^{(1)} + \sqrt{3/2} q_{20}^{(2)} Q_{22}^{(1)} \right) \delta\varphi^-, \quad (47)$$

that describes the oblateness-equatorial ellipticity interactions calls for the same coordinate translation. This retrograde torque gives rise to the Moon’s free libration of latitude corresponding to the TOM subject to the restoring torque (47) (cf. Table 3). Omitting the details, the natural angular frequency is given by $3/2 [(C-A)/B]_M n$ (Rambaux and Williams 2011; cf. Chao 1983), corresponding to a slow motion in the inertial frame with a prograde period of about 24 years, or a retrograde nearly-diurnal period with respect to the 27.3-day lunar rotating frame as would a TOM. The Rambaux and Williams’ (2011) LLR-observed amplitude is rather small at ~ 0.03 arcsecond.

7 Concluding Remarks

In this paper we explored and succeeded in developing the utility of the multipole formalism for two-body gravitational dynamics of various planetary systems and phenomena. The outcomes and lessons learned have been multi-fold.

First, we derived the multipole-expansion formulas for the gravitational field, energy, force, and torque in their exact and concise forms. These formulas, being general, are presented “once and for all” in the sense that they can be manipulated further to suit the specific configuration at hand simply by pertinent coordinate transformations. We demonstrated this for two canonical planetary two-body configurations, namely the “MICG” (needing no coordinate transformation) and “planet+satellite” (needing coordinate translation) types of configurations, understandably often exemplified with the Earth interior or the Earth–Moon system in mind. It should be emphasized that the symmetry, both in physics and mathematics, is a crucial underpinning of and responsible for the generality of the formalism.

Secondly, we have achieved the goal of placing under a single unified theory the various gravitational dynamical phenomena on equal footing, ranging from potential field, energy, satellite orbit, to tide, librations of sorts and precession/nutation, to planetary wobble and rotational and translational normal modes. They follow and fall naturally from the logic of the multipole formalism, that would be otherwise and typically treated topic-by-topic in respective equivalent forms in textbook literature.

Thirdly, in the process we have relied critically upon the powerful mathematics of the complex spherical harmonics developed in QM; the manipulations would otherwise have been rather involved as often seen in the literature of gravitation. In the present case practically the only labor required is of algebraic nature merely involving the normalization conventions and scaling factors. Retaining only the leading terms reduces the multipole expansion to the conventional textbook results, progressively more precise from zeroth-order monopole to axi-symmetric zonal quadrupole to additional quadrupoles that represent the triaxiality, on to even finer subtleties by higher multipoles, all in a systematic manner.

In principle, the multipoles describe the existing density distribution of a 3-D body “as is”, regardless of its state or behavior. As such, the multipole formalism applies most naturally and efficiently to rigid bodies, whereof the multipoles stay constant of time in the body-fixed coordinates. In practice, nonetheless, deformations or mass redistributions of, say, the elastic Earth under forcing can be formulated just the same way, as long as the (time-variable) deformation can be tracked, for instance in terms of Love numbers (themselves also based on spherical harmonics) in the case of the GRACE’s time-variable gravity observations.

The MICG rotational modes and motions discussed in Sect. 5 ought to be contrasted with those derived as eigen-solutions in Mathews et al. (1991, henceforth M91; see also Mathews et al. 2002). Both cases adopt a mantle/outer-core/inner-core three-layer Earth model. M91’s model, like those in other parallel studies on this subject including that adopted in IERS (International Earth Rotation and Reference Systems Service) Convention 2010 (based on Mathews et al. 2002), is axi-symmetric consisting only of the $[0, 0]$ monopole and $[2, 0]$ zonal quadrupoles. By default that excludes the modal solutions that involve other multipoles. In contrast, our Earth model is in general inclusive of all multipoles, a particular consequence of which is the finding of the modes arising with the presence of the $[2, 2]$ quadrupoles for equatorial ellipticities—namely the ATL and the unnamed TOM-like mode (see Table 3). These are degrees of freedom that are absent in the M91 (or similar) model solutions, and should be present in triaxial Earth models such as those discussed in Guo and Shen (2020). On the other hand, of the four rotational modes solved by M91, the free-core nutation and the free inner-core nutation (besides the mantle Chandler wobble and the ICW) arise from the fluid pressure field on the (oblate) core-mantle and inner-core boundaries, and not relying on gravitation for restoring torque (as does ICW). Therefore they do not appear in our MICG solutions in the present paper.

More advanced cases with more complex configurations beyond what we present here can conceivably be treated via the multipole formalism as well. For example, with oblateness somewhat smaller than the mantle’s and subject to the additional MICG torque besides the luni-solar tidal torque, the inner core would precess somewhat differently than the mantle. This differential precession is thus a hybrid MICG + planet-satellite problem, whose dynamics can be formulated by combining what we have demonstrated along with appropriate coordinate rotation. The mantle is subject to the same (but opposite) MICG torque; the resultant (although relatively small) effect can be evaluated the same way. Similarly, the hybrid case of physical libration arises where the satellite itself has an interior MICG configuration, such as the layered Enceladus locked in synchronous rotation with respect to the planet Saturn (Van Hoolst et al. 2008).

Potential extended applications of multipoles can be found with respect to celestial mechanics of orbit perturbations and resonances, spin–orbit coupling (e.g., Murray and Dermott 1999; Naidu and Margot 2015), as well as integrated effects of tidal energy dissipation (cf. Efroimsky and Williams 2009) and angular momentum transports in planetary systems. Shih and Chao (2020) conducted gravitational inversion of the $[2, 2]$

density of the mantle constrained by both the observed Q_{22} and the mantle's q_{22} inferred from a core-dynamic configuration postulation; the results pertain closely to the density anomaly of the prominent LLSVP (Large Low-Shear Velocity Provinces) construct in the lower-mantle.

In summary, the strength of the multipole formalism lies in its clear and concise physics and ease of mathematical manipulation. In this paper we have presented the multipole formalism for the canonical textbook results in planetary two-body dynamics; they are meant to demonstrate the utility of the multipole expansion as a unifying theory. As with the theoretical physics of EM and QM, the multipole formalism is to be carried further in gravitational dynamics problems beyond what are demonstrated here.

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Author contributions SAS is instrumental in the verification and interpretations of all the equations, and initiated the enlargement to planetary dynamic problems.

Data availability The work is theoretical in nature and involves no observational data.

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Declarations

Conflict of interest The author declare no conflict of interest.

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